

THE ELSHOLTZ–TAO TYPE I SUM IS $O(N \log^2 N)$

ANONYMOUS

ABSTRACT. We prove that $\sum_{p \leq N} f_I(p) \ll N \log^2 N$, where $f_I(p)$ counts the Type I solutions of $4/p = 1/x + 1/y + 1/z$. This removes the double logarithm from the upper bound of Elsholtz–Tao, who conjectured that it can be removed; with their lower bound the sum has exact order $N \log^2 N$. The associated CM lifts of discriminant $-4d$ are uniformly separated in $\mathbb{H}$. Sobolev duality and the spectral large sieve give a per- d count complementary to a per- a Poisson–Weil count; their savings degenerate linearly at $d = a$. The exceptional spectrum is controlled on average over the levels $4dq^2$, $d \asymp D$, by Deshouillers–Iwaniec’s Theorem 7 for $a_n = 1$ and its nebentypus form due to Drappeau, after partial summation. To remove the last $\log \log N$ we make both estimates effective, with ε -constant $C_\varepsilon \leq \exp(\exp(A_0/\varepsilon))$; this requires repairs of a few printed details and, for nebentypus, an extra branch of the induction. Using only the published, unquantified form of these estimates one gets $o(N \log^2 N \log \log N)$; under Selberg’s eigenvalue conjecture a shorter proof gives $O(N \log^2 N)$. The results are proved relative to classical inputs listed explicitly (exact trace formulae, Weil-type bounds, Selberg’s $3/16$, and others) and to the inherited Elsholtz–Tao/internal-note reduction. They have been reviewed internally only and need expert scrutiny.

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1. INTRODUCTION

The Erdős–Straus equation is

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad x, y, z \in \mathbb{N}. \quad (1.1)$$

Following Elsholtz–Tao [ET], a solution is of Type I if $n \mid x$ and $(n, yz) = 1$, and of Type II if $n \mid y, z$ and $(n, x) = 1$. The counts retain coordinate order; for odd primes, $f(p) = 3f_I(p) + 3f_{II}(p)$ [ET, (1.4)]. Their Theorem 1.1 states

$$N \log^2 N \ll \sum_{p \leq N} f_I(p) \ll N \log^2 N \log \log N, \quad \sum_{p \leq N} f_{II}(p) \asymp N \log^2 N. \quad (1.2)$$

On p. 5 of arXiv:1107.1010v6, after Theorem 1.1, they attribute the double logarithm to Brun–Titchmarsh in very short progressions and conjecture its elimination. In §9, p. 36, a short modulus avoids the loss for Type II. Their parenthesis nevertheless says no similar trick seems available in the “Type II case” [sic]; reading this as Type I is the interpretation of [MN3, §1], not the literal quotation.

Our main result removes the double logarithm.

Theorem 1.

$$\sum_{p \leq N} f_I(p) \ll N \log^2 N.$$

Hence, by (1.2), $\sum_{p \leq N} f_I(p) \asymp N \log^2 N$.

This is Theorem 9.9(ii). It is PROVED relative to the cited inputs listed under “Status and dependencies” below, which include the classical black boxes (B1)–(B5) of Appendix A:

- (B1) the Petersson and Kuznetsov trace formulae for $\Gamma_0(q)$, for $(\Gamma_0(M), \chi)$ [Dr, Lemma 4.5], and for $\mathrm{SL}_2(\mathbb{Z})$;
- (B2) Weil’s bound for Kloosterman sums;
- (B3) a twisted Weil bound with absolute constants [Dr, Lemma 4.2];
- (B4) Selberg’s eigenvalue bound $\lambda_1 \geq 3/16$ for congruence subgroups;
- (B5) $\lambda_1(\mathrm{SL}_2(\mathbb{Z})) > 1/4$ [DI, Theorem 3].

The proof uses Deshouillers–Iwaniec’s bound [DI, Theorem 7] for the exceptional spectrum averaged over the level, and its nebentypus form [Dr, Lemma 4.10], with a quantified ε -constant $C_\varepsilon \leq \exp(\exp(A_0/\varepsilon))$. Neither paper specifies this dependence. Appendix A proves it (Theorem A.1) by re-deriving the proofs with explicit constants. Two weaker statements need less.

Theorem 2. *Using [DI, Theorem 7] and [Dr, Lemma 4.10] only in their published, unquantified form (and not Appendix A), there is an absolute constant C such that for every $\varepsilon_0 \in (0, 1/4]$*

$$\sum_{p \leq N} f_I(p) \leq C\varepsilon_0 N \log^2 N \log \log N + O_{\varepsilon_0}(N \log^2 N).$$

In particular $\sum_{p \leq N} f_I(p) = o(N \log^2 N \log \log N)$.

This is Theorem 9.9(i); Theorem 9.9(iii) gives the rate for any growth of C_ε . Second, Theorem 1 has a shorter proof under Selberg’s eigenvalue conjecture for the relevant family, which avoids [DI, Theorem 7], [Dr, Lemma 4.10] and the appendix. Put $\Delta = y^2(\partial_x^2 + \partial_y^2)$, so that $-\Delta$ is nonnegative.

Hypothesis 1.1 (SEL). For every $d \geq 1$, every odd squarefree $q \geq 1$ with $(q, d) = 1$, and every even Dirichlet character χ modulo $2q$ (equivalently modulo q), $-\Delta$ on $L^2(\Gamma_0(4dq^2) \backslash \mathbb{H}, \chi)$ has no eigenvalue in $(0, 1/4)$.

Theorem (Theorem 8.2). Under (SEL), $\sum_{p \leq N} f_I(p) \ll N \log^2 N$.

This was the first route (Sections 2–8); the estimates are uniform in levels and characters, and constants are removed before the spectral estimate is applied. None of these results concerns the existence of an Erdős–Straus representation for every prime; they are average upper bounds.

The new input is uniform separation of the level- d CM points of discriminant $-4d$. Sobolev duality then pairs them with a mean-zero Poincaré series at a cost equal to the square root of their number times a second-order Sobolev norm, with no level loss. A spectral large sieve controls that norm, apart from the exceptional spectrum. The fixed- d count complements a fixed- a Weil count precisely at $d = a$; Lemma 8.1 sums the saving even though it degenerates linearly at that boundary. Only absolute errors averaged over d , not uniform per-packet relative errors, are used.

The exceptional spectrum is the only obstacle to removing (SEL). Single-level estimates (Kim–Sarnak, [DI, Theorem 5]) leave a strip of positive width next to $d = a$ (Section 10). The remedy (Section 9) is that the assembly already sums over $d \asymp D$, i.e. over the levels $4dq^2$, and that the unfolded coefficients of a box Poincaré series are smooth in n . Partial summation reduces them to $a_n = 1$, where [DI, Theorem 7] bounds the whole level-averaged exceptional spectrum at the size of the regular spectrum, up to $C_\varepsilon N^{O(\varepsilon)}$. Against the saving $N^{-\delta/4}$, $\delta = 2\alpha - 1 + \gamma$, this leaves untreated only the layers with $\delta < w$, for any fixed $w > 0$; they cost $O(wN \log^2 N \log \log N)$, which gives Theorem 2. The effective bound $C_\varepsilon \leq \exp(\exp(A_0/\varepsilon))$ allows $w \asymp 1/\log \log N$, which gives Theorem 1.

Appendix A proves that bound. Every ε -loss in DI’s proof of Theorem 7 (via their Theorems 2 and 14 and Lemma 8.1) is replaced by an explicit constant. A Gevrey-class cutoff controls the integrations by parts of growing order, and explicit divisor bounds produce the double exponential. The induction on Q [DI, (8.19)] then closes with explicit constants (Proposition A.3). In Drappeau’s nebenspectrum version the switched modulus carries a factor $1/q_0$ and may fall below the induction range; an extra branch handles it. Several minor slips in the printed proofs were found and repaired (Section A.6); none affects the theorems of [DI, Dr].

Status and dependencies. Theorems 1 and 2 are PROVED relative to cited inputs, with no spectral hypothesis, and Theorem 8.2 is CONDITIONAL on (SEL). The cited inputs are the ET reduction (Proposition 2.2, Lemma 2.8, and (8.1)–(8.2)); the internal note [MN3, Proposition 2.3 and proof of Theorem 3.8, step (1)] (MN3’s Proposition 2.3 is Proposition 2.4 here; R114 repair); Deshouillers–Iwaniec [DI, Theorem 2] and Drappeau [Dr, Proposition 4.7, arXiv numbering]; the spectral theorem; Weil’s bound; Brun–Titchmarsh; Selberg’s upper-bound sieve; and Pólya–Vinogradov. Theorems 1 and 2 use in addition [DI, Theorem 7], [Dr, Lemma 4.10] (read as uniform in the character modulus; for Theorem 1 this uniformity is proved in Appendix A), and the absence of nonconstant residual spectrum for congruence subgroups [Iw, Hux]. Theorem 1 also uses Appendix A, which is PROVED relative to (B1)–(B5) and to the derivation files of its source note [TTL3]. Some fixed absolute numerical majorants there were only shape-checked; this affects the numerical value of A_0 , not the result. Appendix A and its source received three internal referee

reports and two independent deep reviews, with no fatal or major findings; it has not been externally refereed. Section 9 (O118) received two internal reviews of its source note and an internal referee pass of the write-up (R118). DI Theorem 5 enters the discussion of exceptional spectrum, not the proofs. The (SEL) argument received internal adversarial review (R111, rounds 1–2), not external refereeing. That review did not re-check the inherited ET/MN3 reduction; MN3 received a separate internal review. (R114 repair) A later internal referee pass (R114) re-derived the parts of that reduction used here, namely $f_I(p) \leq 2 \sum_c w_c(p)$ from [ET, Proposition 2.2, Lemma 2.8] and the large- c step from [ET, (8.1)–(8.2)]; this is still internal checking only. The book locators in [Iw] are inherited from the source note and have not been independently verified; the same applies to the locators in [HR] and to the precise statements of [MV] and [KS], which were not available to the internal reviewers (R114 repair). Computational checks are not proofs. The result needs expert scrutiny before it is relied upon.

Literature and scope. A limited literature search found no later removal of the Type I double logarithm; it was not an exhaustive priority search. Jia [Jia] is cited in ET’s discussion following Theorem 1.1, but for the Type II prime sum; the final arXiv version still calls the Type I double logarithm a technical limitation and conjectures its removal. Duke [Duke] and Liu–Masri–Young [LMY] address related CM distributions, but do not supply the joint level–discriminant estimate needed here, with level comparable to the discriminant.

2. THE TYPE I SUM AND THE BAD REGION

Write $\mathcal{L} = \log N$ and work first with $N/2 < n \leq N$. The nonnegative weight

$$w_c(n) = \#\{(a, d, f) \in \mathbb{N}^3 : f \mid 4a^2d + 1, n = 4acd - f, n/4 < acd \leq 3n/4\} \quad (2.1)$$

satisfies $f_I(p) \leq 2 \sum_c w_c(p)$, by [ET, Proposition 2.2, Lemma 2.8]; the factor two restores the order of y, z . ET’s map is $(x, y, z) = (abdn, acd, bcd)$, with $(abcd, n) = 1$ and $\gcd(a, b, c) = 1$. With $y \leq z$, it has

$$4abd = ne + 1, \quad ce = a + b, \quad 4acd = n + f, \quad ef = 4a^2d + 1, \quad bf = na + c, \quad (2.2)$$

and $a \leq b$, $an \leq bf \leq 5an/3$. Conversely, for every tuple in (2.1), put $e = (4a^2d + 1)/f$ and $b = ce - a$. Direct substitution gives all the identities (2.2), but only $b \geq a/2$, not necessarily $b \geq a$: indeed $0 < f \leq 2n$ and $bf = na + c$. We retain this larger set throughout. Writing $a = N^\alpha$, $b = N^\beta$, $c = N^\gamma$, we have

$$d \asymp N^{1-\alpha-\gamma}, \quad e \asymp N^{\beta-\gamma}, \quad f \asymp N^{1+\alpha-\beta}, \quad \beta \geq \alpha - O(\mathcal{L}^{-1}), \quad \alpha + \gamma \leq 1 + O(\mathcal{L}^{-1}). \quad (2.3)$$

All exponent regions below are understood up to these bounded dyadic errors.

Lemma 2.1 (SL₂ form; [MN3, Lemma 3.1]). *The identity $ef - 4a^2d = 1$ is equivalent to*

$$M = \begin{pmatrix} e & 2a \\ 2ad & f \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \quad n = 2cM_{21} - M_{22}.$$

If $T = \mathrm{diag}(1, d)$, then $\sigma_d(M) = TM^tT^{-1}$ is an anti-involution and $S_d = \{M \in \mathrm{SL}_2(\mathbb{Z}) : M_{21} = dM_{12}\}$ is its fixed set. It is stable under $M \mapsto \gamma M \sigma_d(\gamma)$ for $\gamma \in \Gamma_0(d)$. The positive Type I data also give the positive definite form $[f, 4ad, de]$ of discriminant $-4d$.

Proof. The determinant and linear-form assertions follow by multiplication. For $\gamma = \begin{pmatrix} p & t \\ r & s \end{pmatrix}$, $\sigma_d(\gamma) = \begin{pmatrix} p & r/d \\ dt & s \end{pmatrix}$ is integral exactly when $d \mid r$; reversing products proves stability. The form has discriminant $16a^2d^2 - 4def = -4d$ and positive leading coefficient. Conversely its coefficients recover f, a, e when the middle coefficient is divisible by $4d$. Positivity and this parity restriction are extra conditions, not invariants of all of S_d . $\square$

Remark 2.2 (Single-progression parametrisations). The generic elimination in [MN3, Lemma 3.2] produces the moduli

$$4ad, \quad 4bd, \quad 4ab, \quad 4acf, \quad 4cdf, \quad 4bcf', \quad 4cdf', \quad f' = 4bcd - n.$$

Its claim that these exhaust all three-coordinate single-progression parametrisations has status EVIDENCE: degenerate eliminants were not checked systematically. No completeness claim is used here. The following explicit fibres suffice. Fixing (a, c, f) forces $d \equiv -(4a^2)^{-1} \pmod{f}$ and puts n in one class modulo $4acf$. Fixing (c, d, f) puts a in $\rho_d(f) = \#\{x \pmod{f} : 4dx^2 + 1 \equiv 0 \pmod{f}\}$ classes, hence n in that many classes modulo $4cdf$. Fixing (a, b, c) fixes $e = (a + b)/c$; then $(e, 4ab) = 1$, $d \equiv (4ab)^{-1} \pmod{e}$, and $n = (4abd - 1)/e$ lies in one class modulo $4ab$. Every map from the free variable to n is injective. For fixed (a, b) there are at most $\tau(a + b)$ choices of c . The moduli $4ad, 4bd$ arise by varying c with the appropriate divisor fixed; swapping a, b gives the twins.

Proposition 2.3 (Bad-region geometry; [MN3, Proposition 3.3]). *In the ideal exponent domain $0 \leq \alpha \leq \beta$, $\alpha + \gamma \leq 1$, all seven displayed moduli have exponent at least $1 - \eta$ exactly on*

$$\mathcal{R}_{\text{bad}}(\eta) = \{\gamma \leq \eta, \alpha + \beta \geq 1 - \eta, \beta \leq 2\alpha + \gamma + \eta, \beta \leq 1 + \eta\}. \quad (2.4)$$

Its limiting $\gamma = \eta = 0$ slice has area $1/6$. There e, f are at least $\max(a, d)$ in exponent; for positive η this holds only up to $N^{-O(\eta)}$.

Proof. The seven exponents, in the order of the preceding list, are

$$1 - \gamma, \quad 1 - \gamma + \beta - \alpha, \quad \alpha + \beta, \quad 1 + 2\alpha + \gamma - \beta, \quad 2 - \beta, \quad 1 + 2\beta + \gamma - \alpha, \quad 2 - 2\alpha + \beta.$$

The second and last two inequalities are redundant, giving (2.4). At $\gamma = \eta = 0$ the slice is $\alpha + \beta \geq 1$, $\alpha \leq \beta \leq \min(2\alpha, 1)$, of area $\int_{1/3}^{1/2} (3\alpha - 1) d\alpha + \int_{1/2}^1 (1 - \alpha) d\alpha = 1/6$. Both β and $1 + \alpha - \beta$ are at least $\max(\alpha, 1 - \alpha)$ there. Perturbing the inequalities proves the last assertion. For our enlarged weights $b \geq a/2$ introduces only $O(\mathcal{L}^{-1})$ errors. $\square$

A positive area alone does not prove positive logarithmic mass or exclude other parametrisations. We use the region only to organise an upper bound.

2.1. Arithmetic estimates outside the bad region. We record the imported divisor estimate explicitly. Here $\omega(k)$ is the number of distinct prime divisors.

Proposition 2.4 (Coprimality gain; [MN3, Proposition 2.3]). *Fix $l \geq 1$ and let $k \geq 1$, $U, V \geq 2$. (a) If $kV^2 \leq U^l$ and $U \geq \omega(k) + 2$, then*

$$\sum_{u \leq U, v \leq V} \tau(kuv^2 + 1) \ll_l \frac{\varphi(k)}{k} UV \log U.$$

(b) If $kU \leq V^l$ and $V \geq \omega(2k) + 2$, then the same sum is

$$\ll_l \frac{\varphi(k)}{k} UV \log V \left[1 + \min \left\{ \log(1+k), \frac{k^{1/6} \log(2kU)}{\sqrt{U}} \right\} \right].$$

This is a proved input relative to [ET, Theorem 7.1, Corollary 7.4] and Pólya–Vinogradov. Its proof in [MN3, §2] keeps the indicator that divisors are coprime to k , rather than dropping it in the quadratic root bound. We do not re-prove that imported proposition.

Put $g(x) = x/\varphi(x)$ and write $x \asymp X$ for a dyadic interval $X < x \leq 2X$, allowing a separate initial interval containing 1.

Lemma 2.5 (Weighted divisor sums). *Uniformly in $A, B, D, F, X \geq 1$,*

$$(a) \quad \sum_{x \asymp X} g(x)^2 \ll X, \quad g(x) = \sum_{l|x} \frac{\mu^2(l)}{\varphi(l)}, \quad (2.5)$$

$$(b) \quad \sum_{\substack{a \asymp A, b \asymp B \\ a \leq 2b}} \tau(a+b) g(a) g(b) \ll AB \log(2B), \quad (2.6)$$

$$(c) \quad \sum_{d \asymp D} g(d) \sum_{f \asymp F} \frac{\rho_d(f)}{\varphi(f)} \ll D. \quad (2.7)$$

Proof. The identity in (2.5) follows on prime powers. Also $g^2 = 1 * h$ with $h \geq 0$, supported on squarefree integers, and $h(p) = (p/(p-1))^2 - 1 = O(1/p)$. Thus $\sum_l h(l)/l < \infty$, which proves the mean square and, by Cauchy–Schwarz, $\sum_{x \asymp X} g(x) \ll X$. The same divisor expansion gives $\sum_{x \leq X} 1/\varphi(x) \ll \log(2X)$.

For (2.6), fix a and retain $a \leq 2b$, so $a+b \leq 6B$. The elementary bound $\tau(n) \leq 2 \sum_{\delta|n, \delta \leq \sqrt{n}} 1$ gives, for squarefree l ,

$$\sum_{\substack{b \asymp B, l|b \\ a \leq 2b}} \tau(a+b) \leq 2 \sum_{\delta \leq \sqrt{6B}} \left(\frac{2B(l, \delta)}{l\delta} + 1 \right) \ll \frac{B}{l} \tau(l) \log(2B) + \sqrt{B}.$$

Multiply by $\mu^2(l)/\varphi(l)$ and sum $l \leq 2B$. The series $\sum_l \mu^2(l) \tau(l)/(l\varphi(l))$ converges and the remaining sum is $O(\sqrt{B} \log(2B))$. Summing with weight $g(a)$ proves (2.6).

For (2.7), set $\chi_d = (\frac{-4d}{\cdot})$ and $S_d(Y) = \sum_{r \leq Y} \chi_d(r)/r$, with $S_d(Y) = 0$ if $Y < 1$. This is a nonprincipal real Dirichlet character modulo $4d$. The prime-power root counts give

$$\rho_d(f) \leq (1 * \chi_d)(f), \quad \rho_d(kf) \leq 2^{\omega(k)} \rho_d(f) \quad (k \text{ squarefree}).$$

Indeed even moduli have no roots; for odd $p \nmid d$ the count at every positive power is $1 + \chi_d(p)$, and for $p \mid d$ it is zero. For $Y \geq 1$,

$$\sum_{Y < f \leq 2Y} (1 * \chi_d)(f) = Y S_d(2Y) + O(Y). \quad (2.8)$$

The floor-function error is $O(\sum_{r \leq 2Y} 1) = O(Y)$. We claim, uniformly in $Y \geq 1$,

$$\sum_{d \asymp D} |S_d(Y)|^2 \ll D. \quad (2.9)$$

For $Y \leq Y_0 = D/\log^2(2D)$ expand the square. Terms with rr' even vanish. If rr' is a square the inner d -sum is at most $O(D)$, and $\sum_{rr'=\square} (rr')^{-1} < \infty$ (write

$r = su^2, r' = sv^2$ with s squarefree). Otherwise Pólya–Vinogradov for $d \mapsto (\frac{d}{rr'})$ bounds that sum by $O(\sqrt{rr'} \log(2rr'))$. The total is $O(D + Y \log(2Y)) = O(D)$. For $Y > Y_0$, partial summation and Pólya–Vinogradov modulo $4d$ give

$$S_d(Y) - S_d(Y_0) \ll \frac{\sqrt{D} \log(2D)}{Y_0} = O(1).$$

The last assertion is uniform for large D ; bounded D are handled directly by periodicity and partial summation. This proves (2.9); see [IK, Chapter 12] for the character-sum input. Finally the divisor expansion of $g(f)$, the root inequalities and (2.8) imply

$$\sum_{f \asymp F} \frac{\rho_d(f)}{\varphi(f)} \ll \sum_{k \geq 1} \frac{\mu^2(k) 2^{\omega(k)}}{k \varphi(k)} (|S_d(2F/k)| + 1).$$

When $F/k < 1$ the inner interval contains at most 1, so the same bound holds. Apply Cauchy–Schwarz using (2.5) and (2.9), and sum the convergent k -series. $\square$

Lemma 2.6 (Brun–Titchmarsh outside the bad region). *Fix $0 < \eta_1 < 1/2$. For $P \in \{ab, acf, cdf\}$, the number of tuples in (2.1), with arbitrary c , n prime in $(N/2, N]$, and $m_P \leq N^{1-\eta_1}$, is $O(\eta_1^{-1} N \mathcal{L}^2)$, where $m_{ab} = 4ab$, $m_{acf} = 4acf$, $m_{cdf} = 4cdf$.*

Proof. For $N \geq N_0(\eta_1)$, Brun–Titchmarsh [MV] on an interval of length $N/2$ gives $O(N/(\eta_1 \mathcal{L} \varphi(m)))$ primes per primitive class. A nonprimitive class contains at most one prime. By the injective fibres above, each fixed triple contributes at most

$$\rho_T \left(\frac{2N}{\eta_1 \mathcal{L} \varphi(m_T)} + 1 \right), \quad \rho_T = 1, 1, \rho_d(f),$$

respectively for ab, acf, cdf , with an inessential enlargement of the absolute constant. Use $\varphi(4x) \geq 2\varphi(x)$ and $\varphi(xy) \geq \varphi(x)\varphi(y)$. For acf the reciprocal-totient sum is $O(\mathcal{L}^3)$. For ab it is bounded by $\sum_{ab \leq N, a \leq 2b} \tau(a+b)/(\varphi(a)\varphi(b)) = O(\mathcal{L}^3)$, by (2.6) on $O(\mathcal{L}^2)$ boxes. For cdf , (2.7) gives $\sum_{df \leq N} \rho_d(f)/(\varphi(d)\varphi(f)) = O(\mathcal{L}^2)$; the c -sum supplies one more logarithm. The $+1$ errors total $O_\epsilon(N^{1-\eta_1+\epsilon})$, by the divisor bound and $\rho_d(f) \leq 2^{\omega(f)}$. Choose $\epsilon < \eta_1/2$. $\square$

Lemma 2.7 (Weighted Type I masses). *Let $\mathcal{L}^2 \leq A \leq N$ and $1 \leq D \leq N$. (a1) If $D \leq A$, then $\sum_{a \asymp A, d \asymp D} \tau(4a^2d+1)g(d) \ll AD\mathcal{L}$; if $D \geq A$, the same holds with $g(a)$ in place of $g(d)$. (a2) If also $A, D \geq N^{1/4}$, then*

$$\sum_{a \asymp A, d \asymp D} \tau(4a^2d+1)g(a)g(d) \ll AD\mathcal{L}. \quad (2.10)$$

(b) For every $A, D \geq 1$ and $F \leq 4A$,

$$\sum_{d \asymp D} g(d) \# \{(a, f) : a \asymp A, f \asymp F, f \mid 4a^2d+1\} \ll AD. \quad (2.11)$$

The last assertion also holds for e in place of f .

Proof. Expand $g(d) = \sum_{l \mid d} \mu^2(l)/\varphi(l)$ when $D \leq A$ and put $d = ld'$. Proposition 2.4, second assertion, with coefficient $4l$, linear range $2D/l$, quadratic range $2A$, and

fixed exponent 40, gives

$$\sum_{a \leq 2A, d' \leq 2D/l} \tau(4la^2d' + 1) \ll \frac{AD}{l} \mathcal{L}(1 + \log(1 + 4l)).$$

The required size inequalities are $8D \leq (2A)^{40}$ and $2A \geq \omega(8l) + 2$. If a range is smaller than 2, enlarge it to 2; then $8l \leq (2A)^{40}$ suffices and the loss is bounded since $l \leq 2D$. The resulting l -series with denominator $l\varphi(l)$ converges. For $D \geq A$, expand $g(a)$, put $a = ka'$, and use the first assertion of Proposition 2.4 with coefficient $4k^2$, ranges $2D, 2A/k$. Here $16A^2 \leq (2D)^{40}$ and $2D \geq \omega(4k^2) + 2$; after enlarging a small range use $16k^2 \leq (2D)^{40}$. The bound is $O((AD/k)\mathcal{L})$, summable against $1/\varphi(k)$.

For (2.10), expand both factors and write $a = ka'$, $d = ld'$, with coefficient $K = 4k^2l$. The part $k > A^{1/2}$ or $l > D^{1/2}$ is, for a sufficiently small fixed $\epsilon > 0$,

$$\ll N^\epsilon AD(A^{-1/2} + D^{-1/2}) \ll AD.$$

Here the divisor bound is uniform since $4a^2d + 1 \ll N^3$, and $\sum_{k > T} 1/(k\varphi(k)) \ll T^{-1}$ follows from (2.5). In the remaining part, if $D \geq A$ use the first assertion of Proposition 2.4: its coefficient condition is $16lA^2 \leq (2D/l)^{40}$. If $D < A$ use the second assertion, whose condition is $8k^2D \leq (2A/k)^{40}$. These hold for all sufficiently large N , since the relevant large variable is at least $N^{1/8}$ and the left side is $O(N^3)$. The conditions involving $\omega(K)$ hold as well. Discarding $\varphi(K)/K \leq 1$, both bounds are summable using

$$\sum_{k, l \geq 1} \frac{\mu^2(k)\mu^2(l)(1 + \log(1 + 4k^2l))}{kl\varphi(k)\varphi(l)} < \infty.$$

Finally a fixed (d, f) admits at most $\rho_d(f)(A/f + 1)$ values of a . For $f \leq 2F \leq 8A$ this is at most $9A\rho_d(f)/f$. Equation (2.7) proves (2.11). $\square$

2.2. What remains. Only the following unconditional step of the earlier LD reduction is needed.

Proposition 2.8 (Large- c reduction; [MN3, proof of Theorem 3.8, step (1)]). *For fixed small $\eta > 0$, the contribution of $c > N^\eta$ to $\sum_{N/2 < p \leq N} \sum_c w_c(p)$ is $O(\eta^{-1}N\mathcal{L}^2)$.*

Proof. Here $ad \ll N^{1-\eta}$. Brun–Titchmarsh in $n \equiv -f \pmod{4ad}$ and [ET, (8.1)–(8.2)] on product blocks give

$$\frac{N}{\eta\mathcal{L}} \sum_{ad \ll N} \frac{\tau(4a^2d + 1)}{\varphi(ad)} \ll \eta^{-1}N\mathcal{L}^2.$$

The classes are primitive since $f \mid 4a^2d + 1$ implies $(f, 4ad) = 1$, also for the enlarged weights. $\square$

It remains to count prime tuples with $c \leq N^\eta$ and

$$\min(4ab, 4acf, 4cdf) > N^{1-\eta_1} \quad (\eta_1 \geq \eta). \quad (2.12)$$

The moduli $4ad, 4bd$ are already $\gg N^{1-\eta}$ because $4acd \asymp N$ and $b \geq a/2$; the twin exponents are automatically large. Thus (2.12) lies in $\mathcal{R}_{\text{bad}}(\eta_1)$ up to $O(\mathcal{L}^{-1})$. Smooth upper-bound cells may be supported in $\mathcal{R}_{\text{bad}}(2\eta_1)$, where $e, f \geq N^{1/2-3\eta_1}$ up to absolute constants. This is the only remaining region; in particular no assertion of completeness from Remark 2.2 is needed.

3. $\mathrm{SL}_2/\mathrm{HEEGNER}$ REFORMULATION AND SEPARATION

For $d \geq 1$ define

$$\mathcal{F}_d = \{[A, B, C] \in \mathbb{Z}^3 : B^2 - 4AC = -4d, A > 0, 2d \mid B, d \mid C\}.$$

Its root map is $z_Q = (-B + i\sqrt{4d})/(2A)$. Every form here is positive definite. Type I forms form the subset

$$\mathcal{F}_d^I = \{[f, 4ad, de] : ef - 4a^2d = 1, e, f > 0, a \in \mathbb{Z}\}.$$

The unrestricted integer a makes the invariant set convenient; a positive box will restore $a > 0$. These are CM, or Heegner, forms with discriminant $-4d$ and level d , not forms satisfying the coprime Heegner hypothesis: $(-4d, d) = d > 1$ for $d > 1$ (R114 repair). Type I forms are primitive, since $(f, 4ad) = 1$. The larger set $\mathcal{F}_d$ need not be primitive (e.g. $[2, 6, 6] \in \mathcal{F}_3$); Lemma 6.3 counts only the Type I subset.

Lemma 3.1 (Distance formula). *For positive definite forms Q, Q' of equal discriminant $D < 0$,*

$$\cosh \operatorname{dist}_{\mathbb{H}}(z_Q, z_{Q'}) = 1 + \frac{\operatorname{disc}(Q - Q')}{2|D|}.$$

Proof. Write $Q = [A, B, C]$, $Q' = [A', B', C']$. The formula $\cosh \operatorname{dist}(z, w) - 1 = |z - w|^2/(2\Im z \Im w)$ gives

$$\cosh \operatorname{dist}(z_Q, z_{Q'}) - 1 = \frac{(BA' - B'A)^2 + |D|(A - A')^2}{2|D|AA'}.$$

Substituting $4AC = B^2 + |D|$ and $4A'C' = B'^2 + |D|$ into the discriminant of the difference gives

$$AA' \operatorname{disc}(Q - Q') = (BA' - B'A)^2 + |D|(A - A')^2,$$

which proves the assertion. $\square$

Lemma 3.2 (Uniform separation). *Distinct $Q, Q' \in \mathcal{F}_d$ satisfy $\cosh \operatorname{dist}_{\mathbb{H}}(z_Q, z_{Q'}) \geq 3/2$, independently of d .*

Proof. We have $4d^2 \mid (B - B')^2$ and $4d \mid 4(A - A')(C - C')$, hence $4d \mid \operatorname{disc}(Q - Q')$. By Lemma 3.1, $\cosh \operatorname{dist} - 1 \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. If it is zero, the roots coincide. A positive definite form of given negative discriminant is determined by its root: in $Q = A(X - zY)(X - \bar{z}Y)$ the discriminant fixes $A > 0$. Thus equality zero implies $Q = Q'$, which is excluded. $\square$

Separation persists on every subset, including parity and sieve subsets. It is a statement about all lifts in $\mathbb{H}$, not just a chosen list of representatives on a quotient. This distinction permits the uniform Sobolev argument of §4.

3.1. Coordinates, parity and the sieve group. Set

$$w_Q = z_Q/d = \frac{-2a + i/\sqrt{d}}{f} \quad (Q \in \mathcal{F}_d^I). \quad (3.1)$$

Dilation is a hyperbolic isometry. The set $\mathcal{F}_d$ is preserved by $Q \mapsto Q \circ \gamma$ for $\gamma \in \Gamma^0(d)$, where $\Gamma^0(d)$ means $d \mid \gamma_{12}$. Indeed

$$\begin{aligned} C' &= At^2 + Bts + Cs^2 \equiv 0 \pmod{d}, \\ B' &= 2Apt + B(ps + tr) + 2Crs \equiv 0 \pmod{2d}. \end{aligned}$$

for $\gamma = \begin{pmatrix} p & t \\ r & s \end{pmatrix}$. Roots transform by γ^{-1} . Conjugating by $z = dw$ identifies this with $\Gamma_0(d)$ on the w -side.

The smaller set $\mathcal{F}_d^I$ requires $4d \mid B$. On the z -side it is preserved by $\Gamma^0(2d) \cap \Gamma_0(2)$, since each term in the formula for B' is then divisible by $4d$. On the w -side this becomes $\Gamma_0(2d) \cap \Gamma^0(2)$. In particular, for squarefree odd $(q, d) = 1$, take

$$\Gamma'_q = \Gamma_0(2d) \cap \Gamma(2q), \quad u = w/(2q), \quad M = 4dq^2. \quad (3.2)$$

This preserves both the parity and the condition $q \mid n(Q) = cB - A$. To see the latter, the corresponding z -matrix is congruent to I modulo q , using $(q, d) = 1$; hence it preserves the coefficients of Q modulo q . It is important not to replace this group by $\Gamma_0(d) \cap \Gamma(2)$: for even d that group need not preserve $4d \mid B$ on the w -side.

In u -coordinates the effective group (adjoining $-I$, which acts trivially) is

$$\Gamma_{M,q} = \left\{ \pm \begin{pmatrix} p & t \\ r & s \end{pmatrix} \in \Gamma_0(M) : p \equiv s \equiv 1 \pmod{2q} \right\}.$$

Its cusp at infinity has width one, and it contains $\Gamma_1(M)$. For $q > 1$, its index in $\Gamma_0(M)$ is $\varphi(q)/2$; for $q = 1$ the group is simply $\Gamma_0(4d)$ and the index is 1. The quotient for $q > 1$ is $(\mathbb{Z}/q\mathbb{Z})^\times / \{\pm 1\}$. Consequently the spectral decomposition on $\Gamma_{M,q}$ splits into the even-character spaces on $\Gamma_0(M)$ (with just the trivial space when $q = 1$). This explains precisely the family in (SEL).

3.2. Boxes and Poincaré series. For a smooth compactly supported weight in $a \asymp A, f \asymp F$, its expression on $\mathbb{H}$ is

$$\psi(w) = W_1\left(\frac{-\Re w}{2A\Im w\sqrt{d}}\right) W_2\left(\frac{1}{F\sqrt{d}\Im w}\right).$$

Its height is $\asymp (F\sqrt{d})^{-1}$, width $\asymp A/F$, and hyperbolic area $\asymp A\sqrt{d}$. For spectral estimates use a partition in $(\log f, \log(a/f))$. In the u -coordinate the cells then have the product shape

$$\psi(u) = \phi(x/\lambda)W(y/Y), \quad \lambda \asymp \frac{A}{qF}, \quad Y \asymp \frac{1}{qF\sqrt{d}}, \quad (3.3)$$

where $\text{supp } \phi \subset [-2, -1]$ and $\text{supp } W \subset [1, 2]$. Fixed smooth enlargements cover the original boxes by positivity. In particular $Y/\lambda \asymp (A\sqrt{d})^{-1}$. In the bad region this is a negative power of N , so the condition $Y \leq \lambda L_*^{-3}$ of Proposition 5.1 holds, since its local logarithm $L_* \ll \mathcal{L}$.

Here is the counting identity with the stabilisers made explicit. Let $\tilde{\Lambda}$ be the set of points arising from an invariant subset of $\mathcal{F}_d$ in the chosen coordinate, let Γ be its group, and let its cusp width at infinity be h . Set

$$\psi_{\text{per}}(z) = \sum_{m \in \mathbb{Z}} \psi(z + mh), \quad P_\psi(z) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} \psi_{\text{per}}(\gamma z).$$

Write $\Lambda = \Gamma \backslash \tilde{\Lambda}$ and $e_z = |\Gamma_z / (\Gamma \cap \{\pm I\})|$. This set is finite: there are finitely many integral-form classes of fixed discriminant, and Γ has finite index in the appropriate conjugate of $\text{SL}_2(\mathbb{Z})$. With $d\mu = dx dy/y^2$ and $\#\Lambda = \sum_{z \in \Lambda} e_z^{-1}$, unfolding gives

$$N_d(\psi) := \sum_{w \in \tilde{\Lambda}} \psi(w) = \sum_{w \in \Gamma_\infty \backslash \tilde{\Lambda}} \psi_{\text{per}}(w) = \sum_{z \in \Lambda} \frac{P_\psi(z)}{e_z}. \quad (3.4)$$

Each lift occurs in exactly e_z cosets, cancelled by its weight; root-map injectivity ensures that forms are counted once. Also

$$\langle P_\psi \rangle = \frac{1}{\text{vol}(\Gamma \backslash \mathbb{H})} \int_{\mathbb{H}} \psi d\mu.$$

Periodisation is essential: there is no restriction $\lambda < 1$. The zero Fourier mode when $\lambda > 1$ will contribute a genuine term λ^2/Y to the variance, not an ignorable boundary error.

Finally the Fricke isometry exchanges the two divisor coordinates:

$$-\frac{1}{dw_Q} = \frac{2a + i/\sqrt{d}}{e}.$$

After reflection in the imaginary axis this is the point for $[e, 4ad, df]$. Thus one can use the smaller divisor in a suitable cusp. When e, f are exchanged the original prime functional becomes $cB - C/d$, rather than $cB - A$; the sieve density must be checked for that functional too, as in §6.

4. SOBOLEV DUALITY

Throughout this section Γ denotes the *effective* image in $\text{PSL}_2(\mathbb{R})$ of a finite-index subgroup of $\text{SL}_2(\mathbb{Z})$, or a conjugate thereof. In particular, kernel sums never count both signs of a matrix. Write $V = \text{vol}(\Gamma \backslash \mathbb{H})$. For a Γ -invariant discrete set $\tilde{\Lambda}$ with finitely many orbits, put

$$e_z = |\Gamma_z|, \quad \nu_\Lambda = \sum_{z \in \Gamma \backslash \tilde{\Lambda}} e_z^{-1} \delta_z, \quad \#\Lambda = \sum_{z \in \Gamma \backslash \tilde{\Lambda}} e_z^{-1}.$$

These weights are necessary at elliptic points.

Lemma 4.1 (Resolvent kernel; standard). *There is a nonnegative continuous radial function k_0 on $\mathbb{H}$, with*

$$k_0(r) \ll (1+r)e^{-s_0 r}, \quad s_0 = \frac{1+\sqrt{5}}{2},$$

such that the kernel of $(1-\Delta)^{-2}$ on $L^2(\Gamma \backslash \mathbb{H})$ is $K(z, w) = \sum_{\gamma \in \Gamma} k_0(\text{dist}(z, \gamma w))$. The sum converges absolutely.

Proof. The Selberg–Harish-Chandra multiplier is $h(t) = (5/4 + t^2)^{-2}$; see [Iw, §1.8] for the transform and its inversion, and [Iw, Theorem 1.14] for automorphisation. It is even, holomorphic in $|\Im t| < \sqrt{5}/2$, and integrable against $t \tanh(\pi t) dt$. Inversion therefore gives a continuous bounded point-pair kernel, with $k_0(0) = (4\pi)^{-1} \int_{\mathbb{R}} h(t) t \tanh(\pi t) dt$. For completeness, the decay can also be read directly from the resolvent kernel. For $s > 1$, the kernel of $(-\Delta + s(s-1))^{-1}$ on $\mathbb{H}$ is

$$R_s(r) = \frac{1}{2\pi\sqrt{2}} \int_r^\infty \frac{e^{-(s-1/2)v}}{\sqrt{\cosh v - \cosh r}} dv.$$

Taking the negative derivative in $s(s-1)$, at $s = s_0$, gives

$$k_0(r) = \frac{1}{2\pi\sqrt{2}(2s_0-1)} \int_r^\infty \frac{v e^{-(s_0-1/2)v}}{\sqrt{\cosh v - \cosh r}} dv.$$

This proves positivity. For $r \geq 1$, substitute $v = r + u$ and use $\cosh(r+u) - \cosh r \gg e^r(e^u - 1)$; the resulting integral is $\ll (1+r)e^{-s_0 r}$. At $r = 0$ the numerator cancels the singularity, giving the same boundedness and continuity as inversion. An orbit

of a discrete cofinite group has $O_{z,w,\Gamma}(e^R)$ points, with multiplicity, within distance R . Since $s_0 > 1$, the periodisation is absolutely convergent. The automorphic kernel theorem identifies it with the multiplier h , which is precisely $(1 - \Delta)^{-2}$. $\square$

Lemma 4.2 (Separated sets). *PROVED. If distinct points of $\tilde{\Lambda}$ have distance at least $r_0 > 0$, then*

$$\|\nu_\Lambda\|_{H^{-2}}^2 := \langle \nu_\Lambda, (1 - \Delta)^{-2} \nu_\Lambda \rangle \leq C(r_0) \# \Lambda,$$

where $C(r_0)$ is independent of the group and the set.

Proof. Using Lemma 4.1 and cancelling the stabiliser multiplicity in one variable gives

$$\langle \nu_\Lambda, (1 - \Delta)^{-2} \nu_\Lambda \rangle = \sum_{z \in \Gamma \setminus \tilde{\Lambda}} \frac{1}{e_z} \sum_{w \in \tilde{\Lambda}} k_0(\text{dist}(z, w)).$$

Indeed each lift of w occurs e_w times in the kernel sum. The balls of radius $r_0/2$ around distinct lifts are disjoint. Those with centres within distance R of z lie in the ball of radius $R + r_0/2$; comparison with its area $4\pi \sinh^2((R + r_0/2)/2)$ bounds their number by $O_{r_0}(e^R)$. Thus each inner sum is at most $O_{r_0}(\sum_{m \geq 0} (1 + m)e^{(1-s_0)m}) = O_{r_0}(1)$. All rearrangements are justified by positivity and this convergent bound. $\square$

Proposition 4.3 (Uniform Sobolev duality). *PROVED. Suppose P is smooth and $P, \Delta P, \Delta^2 P \in L^2(\Gamma \setminus \mathbb{H})$. Put $\langle P \rangle = V^{-1} \int P d\mu$ and $P_0 = P - \langle P \rangle$. Under Lemma 4.2's separation hypothesis,*

$$\left| \sum_{z \in \Gamma \setminus \tilde{\Lambda}} \frac{P(z)}{e_z} - \# \Lambda \langle P \rangle \right| \leq C(r_0)^{1/2} (\# \Lambda)^{1/2} \|(1 - \Delta) P_0\|_2.$$

Proof. Lemma 4.2 says that $(1 - \Delta)^{-1} \nu_\Lambda \in L^2$ with squared norm at most $C(r_0) \# \Lambda$. Self-adjointness, first for smooth regularisations of the point masses and then by Sobolev duality, gives

$$\nu_\Lambda(P_0) = \langle (1 - \Delta) P_0, (1 - \Delta)^{-1} \nu_\Lambda \rangle.$$

Point evaluation agrees with this distributional pairing by local Sobolev embedding in dimension two. Cauchy–Schwarz proves the claim. $\square$

Corollary 4.4 (Heegner forms). *PROVED. Let $\mathcal{F} \subset \mathcal{F}_d$ be invariant under a subgroup $\Gamma \subset \Gamma_0(d)$ acting on $w_Q = z_Q/d$, with finitely many orbits. Proposition 4.3 holds with an absolute constant for its root set. For $\psi \in C_c^\infty(\mathbb{H})$, the periodised Poincaré series of §3 satisfies its integrability hypotheses, the counting identity (3.4), and $\langle P_\psi \rangle = V^{-1} \int_{\mathbb{H}} \psi d\mu$. The same holds after conjugating the group by a real dilation.*

Proof. The dilation is a hyperbolic isometry, so Lemma 3.2 supplies the absolute separation $r_0 = \text{arccosh}(3/2)$. Unfold the group sum, cancelling elliptic stabilisers as in Lemma 4.2, and then unfold translations. The root map is injective at fixed discriminant, so this counts forms, not just roots. Unfolding the integral proves the mean formula. The Poincaré series and its Laplacian iterates are smooth and vanish sufficiently high in every cusp: for a nonstabilising coset the transformed height tends to zero, below the support of ψ , whereas at infinity the stabilising coset eventually lies above its support. Thus they satisfy the integrability assumptions of Proposition 4.3. $\square$

5. THE SPECTRAL LARGE SIEVE AND THE VARIANCE OF BOX POINCARÉ SERIES

Use the effective group $\Gamma_{M,q}$ of §3, $M = 4dq^2$, with squarefree $(q, 2d) = 1$ and cusp width one. Write $h_q = [\Gamma_0(M) : \Gamma_{M,q}]$: $h_1 = 1$ and $h_q = \varphi(q)/2$ for $q > 1$. Its h_q character spaces are the even nebentypus spaces modulo q .

The imported spectral estimates. We specify normalisations to distinguish a cusp width from the parameter in the large sieve. At a singular cusp $\mathfrak{a}$, choose a scaling matrix giving period one, and write the expansion of an orthonormal weight-zero Maass form as

$$u_j(\sigma_{\mathfrak{a}}z) = \sqrt{y} \sum_{n \neq 0} \rho_{j\mathfrak{a}}(n) K_{it_j}(2\pi|n|y) e^{2\pi inx}. \quad (5.1)$$

This is DI's normalisation [DI, (1.34), p. 231]. The Eisenstein coefficients $\rho_{\mathfrak{ca}}(n, t)$ will use the identical $\sqrt{y}K_{it}$ convention. The DI cusp parameter for $\Gamma_0(M)$ is

$$\mu(\mathfrak{a}) = \frac{(v, M/v)}{M} \quad \text{if } \mathfrak{a} \sim u/v, v \mid M, (u, v) = 1; \quad \mu(\infty) = M^{-1}.$$

It is *not* the literal translation width at infinity, which is one. For $K \geq 1$, $H \geq 1/2$, and a_n supported on $H < n \leq 2H$, DI [DI, Theorem 2, (1.29)–(1.30), p. 230] bounds the cuspidal expression

$$\sum_{|t_j| \leq K} \frac{1}{\cosh(\pi t_j)} \left| \sum_n a_n \rho_{j\mathfrak{a}}(n) \right|^2$$

and its continuous-spectrum analogue by $(K^2 + \mu(\mathfrak{a})H^{1+\varepsilon}) \sum_n |a_n|^2$, for trivial nebentypus. With an even character of conductor $q_0 \mid q$, the bound is (Drappeau's q_0 is the modulus of χ ; on $\Gamma_0(M)$ the multiplier $\chi(\gamma_{22})$ depends only on the primitive character inducing χ , so q_0 may be taken to be the conductor; R114 repair)

$$\ll_{\varepsilon} (K^2 + q_0^{1/2} \mu(\mathfrak{a})H^{1+\varepsilon}) \sum_n |a_n|^2. \quad (5.2)$$

Here the continuous analogue means $\sum_{\mathfrak{c}} (4\pi)^{-1} \int_{-K}^K \cosh(\pi t)^{-1} |\sum_n a_n \rho_{\mathfrak{ca}}(n, t)|^2 dt$, summed over all singular cusps. This is [Dr, Proposition 4.7, (4.24)–(4.25)], in the arXiv numbering. Drappeau uses Whittaker coefficients: since $W_{0,it}(4\pi|n|y) = 2\sqrt{|n|y}K_{it}(2\pi|n|y)$, $\rho^{\text{DI}}(n) = 2\sqrt{|n|}\rho^{\text{Dr}}(n)$. Thus his $\sqrt{n}\rho^{\text{Dr}}(\pm n)$ is exactly our coefficient up to the constant 2; his factor $(1 + |t|)^{\pm\kappa}$ is one because $\kappa = 0$. These estimates hold separately for either sign of n .

For later comparison, DI [DI, Theorem 5, (1.38), p. 232] states, for trivial character, $X \geq 1$, and exceptional parameters $t_j = i\sigma_j$, $0 < \sigma_j < 1/2$,

$$\sum_{j \text{ exc}} X^{2\sigma_j} \left| \sum_{H < n \leq 2H} a_n \rho_{j\mathfrak{a}}(n) \right|^2 \ll_{\varepsilon} (1 + \sqrt{\mu(\mathfrak{a})HX})(1 + \sqrt{\mu(\mathfrak{a})H^{1+\varepsilon}}) \|a\|_2^2. \quad (5.3)$$

The nebentypus analogue is [Dr, Lemma 4.8], with second factor $1 + (q_0\mu(\mathfrak{a})H)^{1/2+\varepsilon}$. The displayed DI statements have been checked against the scan; the Drappeau numbering and coefficient conversion refer to arXiv:1504.05549, not to an unchecked published numbering.

The box variance. Take $\phi \in C_c^\infty([-2, -1])$, $W \in C_c^\infty([1, 2])$, and

$$\psi(u) = \phi(x/\lambda)W(y/Y), \quad L_* = \log(2 + (\lambda Y)^{-1} + Y^{-1} + M).$$

Assume $\lambda > 0$ and $Y \leq \lambda L_*^{-3}$. Periodisation is always understood; no restriction $\lambda < 1$ is imposed. For nonnegative nonzero weights its area is $\asymp \lambda/Y$. In applications $\lambda \asymp A/(qF)$, $Y \asymp 1/(qF\sqrt{d})$, and $L_* \ll \mathcal{L}$.

Proposition 5.1 (Box variance). *CONDITIONAL on (SEL). On $\Gamma_{M,q} \backslash \mathbb{H}$,*

$$\|(1 - \Delta)(P_\psi - \langle P_\psi \rangle)\|_2^2 \ll_{\phi, W, \varepsilon} L_*^C \left\{ \frac{\lambda}{Y} \left(1 + \frac{q^{1/2}}{M} (L_*/\lambda)^{1+\varepsilon} \right) + \frac{\lambda^2}{Y} \right\}. \quad (5.4)$$

The constant is uniform in d, q, λ, Y , and depends on only finitely many seminorms of the two fixed weights.

Proof. Reduction and spectral expansion. Differentiation commutes with periodisation and

$$(1 - \Delta)\psi = \phi(x/\lambda)(W(v) - v^2 W''(v)) - (Y/\lambda)^2 \phi''(x/\lambda) v^2 W(v), \quad v = y/Y.$$

Moreover $\int \Delta \psi d\mu = 0$. Since $Y/\lambda \leq 1$, it suffices to bound $\|P_\psi - \langle P_\psi \rangle\|_2^2$ by the right side of (5.4), for arbitrary fixed smooth weights of this type. For each even $\chi \bmod q$, let

$$P_\chi = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma_0(M)} \overline{\chi(\gamma)} \psi_{\text{per}}(\gamma u).$$

Character orthogonality gives $P_\psi = h_q^{-1} \sum_\chi P_\chi$ and

$$\|P_\psi - \langle P_\psi \rangle\|_{\Gamma_{M,q}}^2 = \frac{1}{h_q} \sum_\chi \|P_\chi - \mathbf{1}_{\chi=1} \langle P_1 \rangle\|_{\Gamma_0(M), \chi}^2. \quad (5.5)$$

The spectral expansion, with Eisenstein measure $dt/(4\pi)$, is the usual Parseval formula [Iw, Theorem 7.3] (with the character spaces and singular cusps as in [Dr, §4.1]). *This is the sole use of (SEL):* after removing the constant, all discrete parameters are real. (There is no nonconstant residual spectrum in any case; see Section 9.) Thus the squared norm for a fixed character is the sum of $|\langle P_\chi, u_j \rangle|^2$ over cusp forms and $(4\pi)^{-1} \sum_c \int_{\mathbb{R}} |\langle P_\chi, E_c(\cdot, 1/2 + it) \rangle|^2 dt$.

Unfolding and Mellin inversion. Write $\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(v) e^{-2\pi i \xi v} dv$. The periodised box has Fourier coefficient $\lambda \hat{\phi}(\lambda n) W(y/Y)$, even when $\lambda > 1$. Using (5.1), unfolding gives

$$\langle P_\chi, u_j \rangle = \sum_{n \neq 0} \overline{\rho_j(n)} \lambda \hat{\phi}(\lambda n) I_{t_j}(n), \quad I_t(n) = \int_0^\infty W(y/Y) y^{-3/2} K_{it}(2\pi|n|y) dy.$$

Set

$$G_t(s) = \Gamma((s + it)/2) \Gamma((s - it)/2), \quad \mathcal{W}(s) = \int_0^\infty W(v) v^{-3/2-s} dv.$$

Write also

$$B_j(s) = \sum_{n \neq 0} \overline{\rho_j(n)} \lambda \hat{\phi}(\lambda n) |n|^{-s}.$$

The function $\mathcal{W}$ is entire and rapidly decreasing vertically, uniformly on fixed strips. Mellin inversion of the Bessel function gives, for $\sigma > 0$,

$$\langle P_\chi, u_j \rangle = \frac{Y^{-1/2}}{8\pi i} \int_{(\sigma)} G_{t_j}(s) \mathcal{W}(s) (\pi Y)^{-s} B_j(s) ds. \quad (5.6)$$

Absolute convergence follows from the rapid decay of $\widehat{\phi}$, the polynomial growth of Fourier coefficients, and Stirling's formula.

Uniform spectral bounds. Put $\sigma_0 = 1/L_*$. Stirling's formula, including the simple poles of Γ , gives

$$\begin{aligned} |G_t(\sigma_0 + iv)| &\ll L_*^2 e^{-\pi|v|/2} & (|t| \leq 1), \\ |G_t(-1 + iv)| &\ll e^{-\pi|t|/2} (1 + |t|)^{-2} (1 + |v|)^4 & (|t| \geq 1). \end{aligned}$$

For the second bound, if $|v| \leq |t|/2$ both gamma arguments have imaginary part comparable to $|t|$; otherwise $1 + |t| \ll 1 + |v|$ and the exponential is $e^{-\pi \max(|v|, |t|)/2}$. On this line both real parts are $-1/2$, a fixed distance from gamma poles. Since $Y^{-\sigma_0} \ll 1$, Cauchy-Schwarz in (5.6) yields, for $|t_j| \leq 1$,

$$|\langle P_\chi, u_j \rangle|^2 \ll Y^{-1} L_*^4 \int_{\mathbb{R}} |\mathcal{W}(\sigma_0 + iv)| |B_j(\sigma_0 + iv)|^2 dv.$$

For $|t_j| \geq 1$, shift to $\Re s = -1$, crossing only $s = \pm it_j$. The residue contribution is

$$\frac{Y^{-1/2}}{2} \sum_{\pm} \Gamma(\pm it_j) \mathcal{W}(\pm it_j) (\pi Y)^{\mp it_j} B_j(\pm it_j).$$

Let $\widetilde{B}_j(s) = (\pi Y)^{-s} B_j(s)$. Using $|\Gamma(it)|^2 = \pi/(|t| \sinh(\pi|t|))$, for any fixed large B the squared coefficient is at most

$$\begin{aligned} \frac{C_B Y^{-1}}{\cosh(\pi t_j)} &\left\{ (1 + |t_j|)^{-B} \sum_{\pm} |B_j(\pm it_j)|^2 \right. \\ &\quad \left. + (1 + |t_j|)^{-4} \int_{\mathbb{R}} |\mathcal{W}(-1 + iv)| (1 + |v|)^4 |\widetilde{B}_j(-1 + iv)|^2 dv \right\}. \end{aligned} \quad (5.7)$$

The large sieve, including both tails. Split n by sign and into dyadic blocks $H < |n| \leq 2H$, $H = 2^{i-1}$, $i \geq 0$. With $w_H = 1 + |\log(\lambda H)|$, one has $\sum_H w_H^{-2} \ll 1$. Cauchy-Schwarz across blocks and (5.2) imply

$$\mathcal{S}(K; b) := \sum_{|t_j| \leq K} \frac{|\sum_{n \neq 0} \overline{\rho_j(n)} b_n|^2}{\cosh(\pi t_j)} \ll_{\varepsilon} \sum_H w_H^2 (K^2 + q^{1/2} M^{-1} H^{1+\varepsilon}) \|b^{(H)}\|_2^2.$$

Complex conjugation of the coefficient vector causes no change in this inequality; either sign is covered by Drappeau's statement (or reflection and replacement of χ by $\bar{\chi}$). For $b_n(s) = \lambda \widehat{\phi}(\lambda n) |n|^{-s}$, at $\Re s = \sigma_0$ or 0, the block norm is $\ll_B \lambda^2 H (1 + \lambda H)^{-B}$. Summing the geometric tails gives, with $Z = q^{1/2} M^{-1} \lambda^{-1-\varepsilon}$,

$$\mathcal{S}(K; b(s)) \ll L_*^2 \lambda (K^2 + Z). \quad (5.8)$$

At $\Re s = -1$, the coefficients of $\widetilde{B}$ have modulus $\pi Y |n| \lambda |\widehat{\phi}(\lambda n)|$; their bound is (5.8) multiplied by $(Y/\lambda)^2 \leq 1$, after increasing the decay exponent. This also proves the claims when $\lambda > 1$: then every nonzero mode lies in the rapidly decreasing tail.

For the varying vectors $b_n(\pm it_j)$, on each unit interval $[k, k+1]$ use the elementary inequality

$$|B_j(\pm it)|^2 \leq 2|B_j(\pm ik)|^2 + 2 \int_k^{k+1} |\partial_\tau B_j(\pm i\tau)|^2 d\tau.$$

Differentiating multiplies a coefficient by $\log |n|$, bounded on a block by $O(L_* + |\log(\lambda H)|)$; the extra factors are absorbed in the same block sums. Consequently the sum on $K < |t_j| \leq 2K$ for these vectors is $\ll K L_*^4 \lambda (K^2 + Z)$. Insert these

bounds in (5.7) and the small- t estimate. Dyadic summation over $K \geq 1$ converges, since the integral term has weight K^{-4} and the residue term has arbitrary weight K^{-B} . Vertical integrals converge by the decay of $\mathcal{W}$. We obtain

$$\sum_j |\langle P_\chi, u_j \rangle|^2 \ll L_*^C \frac{\lambda}{Y} (1 + Z).$$

No truncation at logarithmic frequency or height was made.

The continuous spectrum and its zero mode. For nonzero modes the identical argument uses the continuous version of (5.2). In the unit-interval argument freeze the Eisenstein parameter in the Fourier coefficients and differentiate only the auxiliary variable in $|n|^{-i\tau}$, then evaluate on the diagonal $\tau = t$; no derivative of an Eisenstein coefficient is required. The constant term at infinity is $\delta_{\infty} y^{1/2+it} + \Phi_{\infty}(1/2 + it) y^{1/2-it}$. Its pairing with the box is

$$\lambda \widehat{\phi}(0) Y^{-1/2} \{ \delta_{\infty} Y^{-it} \mathcal{W}(it) + \overline{\Phi_{\infty}(1/2 + it)} Y^{it} \mathcal{W}(-it) \}.$$

Unitarity of the scattering matrix on the unitary axis gives $\sum_{\mathfrak{c}} |\Phi_{\infty}(1/2 + it)|^2 = 1$. Squaring, integrating and summing therefore costs $O(\lambda^2/Y)$, independently of the number of cusps. The cross term with nonzero modes is absorbed by $|a + b|^2 \leq 2|a|^2 + 2|b|^2$. Finally (5.5) averages exactly h_q character bounds; there is no loss from their number. Enlarging $\lambda^{-1-\varepsilon}$ to $(L_*/\lambda)^{1+\varepsilon}$ proves (5.4). $\square$

Without (SEL), the terms with $t_j = i\sigma_j$ remain. On a block $|n| \asymp H$, the Bessel factor has additional size $(HY)^{-\sigma_j}$. Thus (5.3) is used with $X = (HY)^{-1}$ when $HY \leq 1$, not with $X = Y^{-1}$. For a single d it does not recover Proposition 5.1 in the required cells (Section 10). On average over $d \asymp D$ the exceptional part is bounded unconditionally in Section 9 (Proposition 9.4). Neither the Sobolev argument nor the regular-spectrum large sieve depends on (SEL).

6. PER- d COUNTS

Fix $c \geq 1$ and use $\mathcal{F}_d^I, \Gamma'_q$ from §3, with $n(Q) = cB - A = 4acd - f$. In $[A, B, C]$, A is a form coefficient, not the dyadic parameter; the parity and sieve invariance were proved there.

For a prime $\ell \nmid 2d$, put $\chi_d(\ell) = (-d/\ell)$ (this agrees with $(-4d/\ell)$ of Lemma 2.5; R114 repair) and define $g_{c,d}(\ell)$ as the proportion on the quadric $B^2 - 4AC = -4d$ satisfying $cB - A = 0$. Directly,

$$g_{c,d}(\ell) = \begin{cases} (\ell - 1)/(\ell^2 + \chi_d(\ell)\ell), & \ell \nmid c, \\ (1 + \chi_d(\ell))/(\ell + \chi_d(\ell)), & \ell \mid c. \end{cases} \quad (6.1)$$

Extend multiplicatively to squarefree q coprime to $2d$. The denominator counts the quadric: for $A \neq 0$ there are $\ell(\ell - 1)$ choices, while $A = 0$ gives $\ell(1 + \chi_d(\ell))$. If $\ell \nmid c$, the imposed equation forces $B \neq 0$, giving exactly $\ell - 1$ choices. If $\ell \mid c$, it forces $A = 0$, giving the second numerator. Primes dividing $2d$ never divide n : both f and n are odd, and f is a unit at every prime dividing d .

Lemma 6.1 (Factorisation of the orbital main term). *PROVED. Let $\#\Lambda_d(q)$ be the orbifold count of $\{Q \in \mathcal{F}_d^I : q \mid n(Q)\}$ modulo Γ'_q , and put $V(q) = \text{vol}(\Gamma'_q \backslash \mathbb{H})$. For squarefree $(q, 2d) = 1$,*

$$\frac{\#\Lambda_d(q)}{V(q)} = g_{c,d}(q) \frac{\#\Lambda_d(1)}{V(1)}.$$

Proof. There is nothing to prove for $q = 1$. Reduction of Γ'_1 onto $\mathrm{SL}_2(\mathbb{Z}/q\mathbb{Z})$ is surjective: it contains $\Gamma(2d)$, and the assertion follows from CRT and surjectivity of reduction for $\mathrm{SL}_2(\mathbb{Z})$. The kernel is Γ'_q . Thus the index of effective groups is $I_q = |\mathrm{SL}_2(\mathbb{Z}/q\mathbb{Z})|/2$, since $-I$ belongs to Γ'_1 but not to Γ'_q . At a prime $\ell \mid q$, the stabiliser of a nondegenerate binary form in $\mathrm{SL}_2(\mathbb{F}_\ell)$ is its special orthogonal torus, of order $\ell - \chi_d(\ell)$: it is either the split torus or the norm-one subgroup of $\mathbb{F}_{\ell^2}^\times$. Its orbit consequently has $\ell^2 + \chi_d(\ell)\ell$ elements, the size of the entire quadric. The fraction satisfying the sieve condition is therefore $g_{c,d}(\ell)$, independently of the original orbit. CRT gives the product of these fractions. For an orbit with stabiliser order e_Q , the weighted number of sieved Γ'_q -orbits above it is $e_Q^{-1} I_q g_{c,d}(q)$, by orbit-stabiliser (equivalently, sum over the cosets before dividing by the stabiliser). Summing over the original orbits, and using $V(q) = I_q V(1)$, proves the formula. $\square$

Choose a fixed physical box, so that in the coordinates $u = w/(2q)$ its weights are $\psi_q(u) = \psi_1(qu)$. Accordingly $\lambda_q = \lambda_1/q$, $Y_q = Y_1/q$, and $\int \psi_q d\mu = \int \psi_1 d\mu$. Define

$$\mathfrak{M}_d = \frac{\#\Lambda_d(1)}{V(1)} \int_{\mathbb{H}} \psi_q d\mu, \quad \kappa_c(q) = 2^{\omega((q,c))}.$$

The first quantity is independent of the sieve modulus. Here and below $\omega(m)$ counts the distinct prime factors of m .

Theorem 6.2 (Per- d count). *CONDITIONAL on (SEL). Let q be squarefree, $(q, 2d) = 1$, and let ψ_q be a box as in Proposition 5.1, with $\lambda_q \asymp A/(qF)$, $Y_q \asymp 1/(qF\sqrt{d})$, $A, F \geq 1$. Assume explicitly $Y_q \leq \lambda_q L_*^{-3}$ and polynomial parameter ranges in N , so $L_* \ll \mathcal{L}$. Then, for $0 < \varepsilon < 1/2$,*

$$\left| \sum_{\substack{Q \in \mathcal{F}_d^I \\ q|n(Q)}} \psi_q(u_Q) - g_{c,d}(q) \mathfrak{M}_d \right| \ll_\varepsilon \mathcal{L}^C q \kappa_c(q)^{1/2} (\#\Lambda_d(1))^{1/2} \left\{ A\sqrt{d} \left(1 + \frac{A}{qF} \right) + \frac{F^{1+\varepsilon}}{\sqrt{d}} \right\}^{1/2}. \quad (6.2)$$

In particular the factor $\kappa_c(q)^{1/2}$ is absent when $(q, c) = 1$, as in the sieve application. There is no lower bound on F/A .

Proof. Corollary 4.4 applies to the sieved subset, because separation is preserved by passage to a subset and by the dilation to u . Lemma 6.1 identifies its mean as $g_{c,d}(q) \mathfrak{M}_d$. In Proposition 5.1,

$$\frac{\lambda}{Y} \asymp A\sqrt{d}, \quad \frac{\lambda^2}{Y} \asymp A\sqrt{d} \frac{A}{qF}, \quad \frac{q^{1/2}}{MY} \lambda^{-\varepsilon} \ll \frac{F^{1+\varepsilon}}{\sqrt{d}} q^{-1/2+\varepsilon} A^{-\varepsilon} \leq \frac{F^{1+\varepsilon}}{\sqrt{d}}.$$

The remaining factor from Sobolev duality is $(\#\Lambda_d(q))^{1/2}$. For $\ell \nmid c$,

$$|\mathrm{SL}_2(\mathbb{F}_\ell)| g_{c,d}(\ell) = (\ell - \chi_d(\ell))(\ell - 1) \leq \ell^2;$$

for $\ell \mid c$ it is either zero or $2\ell(\ell - 1) \leq 2\ell^2$. The exact identity from Lemma 6.1 therefore gives $\#\Lambda_d(q) \leq \kappa_c(q) q^2 \#\Lambda_d(1)$, also for $q = 1$. Taking square roots proves (6.2). $\square$

The term $A/(qF)$ is the cost of wrapping around several periods; it carries no N^ε loss. One may instead use the e -cusp: $w \mapsto -1/(dw)$ sends the point to $(2a + i/\sqrt{d})/e$, and reflection gives the same boxes. Interchanging e, f replaces the sieve equation $A = cB$ by $C/d = cB$. The involution $(A, B, C) \mapsto (C/d, B, dA)$ is a bijection of the quadric modulo every prime not dividing $2d$; hence the densities and the estimate are unchanged.

Lemma 6.3 (Class-number bound). *PROVED. With $h(-4d)$ the number of primitive proper classes of discriminant $-4d$, and $r(d) = \prod_{p^k \parallel d} p^{\lfloor k/2 \rfloor}$,*

$$\#\Lambda_d(1) \leq 6h(-4d)r(d), \quad \sum_{d \leq D} \#\Lambda_d(1) \ll D^{3/2} \log^2(2D).$$

Proof. Every Type I form is primitive: a prime dividing f divides neither $2ad$ nor $4ad$, by $ef - 4a^2d = 1$. For a primitive class represented by Q_0 , its admissible level- d cosets are bounded by the number of $(t : s) \in \mathbb{P}^1(\mathbb{Z}/d\mathbb{Z})$ with $Q_0(t, s) \equiv 0 \pmod{d}$. Indeed, the second column of a matrix representing a coset modulo $\Gamma^0(d)$ gives this projective point; the condition that the transformed last coefficient be divisible by d is necessary. Passing from $\Gamma_0(d)$ to Γ'_1 costs index at most six (exactly six for odd d , four for even d); imposing Type I parity can only decrease the count. Stabiliser weights only improve this upper bound.

For $p^k \parallel d$, make an invertible change of variables so that $Q_0 = [A_0, B_0, C_0]$ has $p \nmid A_0$. This is possible by primitivity; at 2, B_0 is even and one of the outer coefficients is odd. Points with $p \mid s$ cannot be zeros. At an odd prime, zeros $(t : 1)$ satisfy $(2A_0t + B_0)^2 \equiv 0 \pmod{p^k}$, since the discriminant is $-4d$. There are exactly $p^{\lfloor k/2 \rfloor}$ possibilities. At 2, write $B_0 = 2B'_0$; the identity

$$A_0Q_0(t, 1) = (A_0t + B'_0)^2 + d$$

shows that the condition is $A_0t + B'_0 \equiv 0 \pmod{2^{\lfloor k/2 \rfloor}}$, again giving $2^{\lfloor k/2 \rfloor}$ possibilities. CRT proves the first assertion. The classical imaginary-quadratic class-number formula [IK] is $h(D_0) = w(D_0)\sqrt{|D_0|}L(1, \chi_{D_0})/(2\pi)$, where $w(D_0) \leq 6$ and χ_{D_0} is the Kronecker character, possibly imprimitive. Its nonprincipal character sums are bounded by $|D_0|$; splitting the Dirichlet series there and using partial summation gives $L(1, \chi_{D_0}) \ll \log(2|D_0|)$. Thus $h(-4d) \ll \sqrt{d} \log(2d)$, including nonfundamental discriminants. Finally,

$$\sum_{d \leq D} r(d) \leq \sum_{m \leq \sqrt{D}} m \#\{d \leq D : m^2 \mid d\} \leq D \sum_{m \leq \sqrt{D}} \frac{1}{m} \ll D \log(2D).$$

Multiplying by $\sqrt{D} \log(2D)$ proves the second assertion. $\square$

There is no uniform per- d relative-error assertion: the orbital main term contains a possibly small class number. Section 8 uses only absolute errors and $\sum_{d \lesssim D} \sqrt{\#\Lambda_d(1)} \ll D^{5/4} \log(2D)$, by Cauchy–Schwarz and Lemma 6.3. Relative to the box mass AD , the first summed error in (6.2) has size $\mathcal{L}^C q(D/A)^{1/2}(1 + A/(qF))^{1/2}$ when $(q, c) = 1$. This is an averaged absolute-error calculation, not a lower bound for an individual $\mathfrak{M}_d$.

7. THE PER- a WEIL COUNT

The complementary estimate is unconditional. It counts a modular hyperbola rather than a packet of forms. Write $e(t) = \exp(2\pi it)$ in this section, and put

$$S(h, k; m) = \sum_{x \bmod m}^* e((hx + k\bar{x})/m).$$

All smooth weights below have fixed derivative bounds. For upper bounds we take them nonnegative, with positive integral.

Proposition 7.1 (the count (K_a) ; PROVED). *Let $a, c \geq 1$, let q be squarefree with $(q, 2a) = 1$, and set $m = 4a^2$, $r = mq$. For $D, E \geq 1$ put $F = mD/E$, and let $W_1, W_2 \in C_c^\infty((1, 2))$. Define*

$$S_a(q) = \sum_{\substack{e, f \geq 1 \\ ef \equiv 1 \pmod{m} \\ q | 4acd - f}} W_1(e/E) W_2(d/D), \quad d = (ef - 1)/m.$$

Then $S_a(q) = g'_{c,a}(q) X_a + R_a(q)$, where

$$X_a = \frac{\varphi(m)}{m^2} \iint_{\mathbb{R}^2} W_1(x/E) W_2((xy - 1)/(mD)) dx dy \asymp D \frac{\varphi(m)}{m}, \quad (7.1)$$

$$|R_a(q)| \ll 2^{\omega(q)} \tau(m)^2 \left(qa + \frac{D}{E} + \frac{D}{F} \right), \quad (7.2)$$

and $g'_{c,a}$ is multiplicative on squarefree numbers, with

$$g'_{c,a}(\ell) = \begin{cases} (\ell - 1)/\ell^2, & \ell \nmid c, \\ 1/\ell, & \ell \mid c. \end{cases}$$

The constants depend only on the weights.

Proof. The identity

$$an = f(ce - a) - c$$

shows that the residue conditions on (e, f) modulo r are the CRT product of

$$\mathcal{S}_m = \{(e, f) : ef = 1 \pmod{m}\}, \quad \mathcal{S}_q = \{(e, f) : f(ce - a) = c \pmod{q}\}.$$

Put $W(x, y) = W_1(x/E) W_2((xy - 1)/(mD))$. On its support $x \asymp E$, $y \asymp F$, and $\partial_x^i \partial_y^j W \ll_{i,j} E^{-i} F^{-j}$. Two-dimensional Poisson summation gives

$$S_a(q) = r^{-2} \sum_{h, k \in \mathbb{Z}} \widehat{W}(h/r, k/r) \mathcal{K}(h, k), \quad (7.3)$$

$$|\widehat{W}(h/r, k/r)| \ll_B EF(1 + E|h|/r)^{-B} (1 + F|k|/r)^{-B}.$$

The sign in the finite Fourier transform is immaterial; with the positive sign, CRT gives

$$\mathcal{K}(h, k) = S(\bar{q}h, \bar{q}k; m) T_q(\bar{m}h, \bar{m}k), \quad T_q(u, v) = \sum_{(e, f) \in \mathcal{S}_q} e((ue + vf)/q).$$

The prime factors of T_q have CRT-rescaled arguments. For a prime $\ell \mid q$, if $\ell \nmid c$, the substitution $w = ce - a$ gives

$$T_\ell(u, v) = e(u\bar{c}a/\ell) S(u\bar{c}, vc; \ell).$$

If $\ell \mid c$, then $f = 0$ and $T_\ell(u, v) = \ell \mathbf{1}_{\ell|u}$. In particular $|T_q(u, v)| \leq 2^{\omega(q)} q$ (the trivial cardinality bound suffices). At $(h, k) = (0, 0)$ we have $|\mathcal{S}_m| = \varphi(m)$ and $|\mathcal{S}_\ell| = \ell - 1$ or ℓ , respectively. This gives exactly (7.1) times $g'_{c,a}(q)$.

We retain the axial frequencies: they are important if a divisor is small. For $B > 2$ and every integer $v \geq 1$,

$$\sum_{\substack{k \neq 0 \\ v|k}} (1 + F|k|/r)^{-B} \ll_B \frac{r}{vF}. \quad (7.4)$$

Indeed, if $vF \leq r$, compare with an integral; otherwise the left side is $O_B((r/(vF))^B)$. Since $|S(0, \bar{q}k; m)| = |c_m(k)| \leq (k, m)$, expanding $(k, m) \leq \sum_{v|(k, m)} v$ in (7.4) gives

$$\sum_{k \neq 0} (k, m) (1 + F|k|/r)^{-B} \ll \tau(m) r/F.$$

Thus the frequencies $h = 0 \neq k$ contribute at most $2^{\omega(q)} \tau(m) E/m = 2^{\omega(q)} \tau(m) D/F$; $k = 0 \neq h$ contributes at most $2^{\omega(q)} \tau(m) D/E$.

For $hk \neq 0$ use the classical Weil–prime-power bound [Weil], in the convenient form

$$|S(u, v; m)| \leq 3\tau(m) m^{1/2} (u, v, m)^{1/2}.$$

The constant allows the prime 2. Expanding the gcd as before yields

$$\begin{aligned} \sum_{h, k \neq 0} (h, k, m)^{1/2} (1 + E|h|/r)^{-B} (1 + F|k|/r)^{-B} \\ \leq \sum_{v|m} v^{1/2} O_B \left(\frac{r}{vE} \frac{r}{vF} \right) \ll_B \frac{r^2}{EF}, \end{aligned}$$

since $\sum_{v \geq 1} v^{-3/2} < \infty$. This remains valid when $vE > r$ or $vF > r$. Inserting it in (7.3) bounds the nonaxial contribution by $O(2^{\omega(q)} q \tau(m) m^{1/2})$. As $m^{1/2} = 2a$, the claimed error follows, even with one power of $\tau(m)$ here. $\square$

For later use, if $\min(E, F) \geq N^{c_0}$, summation over $a \asymp A$ and squarefree $q \leq Q$ gives

$$\sum_{a \asymp A} \sum_{\substack{q \leq Q \\ (q, 2a)=1}} \mu^2(q) 3^{\omega(q)} |R_a(q)| \ll \mathcal{L}^C (Q^2 A^2 + QADN^{-c_0}). \quad (7.5)$$

Here we used the elementary divisor-moment bound $\sum_{a \asymp A} \tau(4a^2)^2 \ll A(\log(2A))^C$. Measured against the scale AD , the error is $\mathcal{L}^C (Q^2 A/D + QN^{-c_0})$. This gives a useful sieve level on the side $D > A$. The axial errors explain why a uniform density assertion with no lower bound on E, F would be false. No small-divisor substitute for this proposition is used below.

8. ASSEMBLY

We begin with the elementary accounting fact that permits a sieve level to approach 1 at the boundary $A = D$.

Lemma 8.1 (linearly degenerating savings; PROVED). *Let $L \geq 2$, and let $m_{jk} \geq 0$, $1 \leq j, k \leq \lfloor L \rfloor$, with $\sum_k m_{jk} \leq \mathcal{M}$ for each j . Then*

$$\sum_{j, k} m_{jk} \min(1/j, 1/k) \leq \mathcal{M}(1 + \log L).$$

If also $m_{jk} \leq \mathcal{M}/L$, the same sum is at most $2\mathcal{M}$. (R114 repair: $\mathcal{M}$ is a mass bound, not the level $M = 4dq^2$.)

Proof. The first assertion follows by using $1/j$ and summing the harmonic series. For the second, group pairs by $h = \max(j, k)$:

$$\sum_{j,k \leq L} \min(1/j, 1/k) = \sum_{h \leq L} \frac{2h-1}{h} \leq 2L.$$

Multiply by $\mathcal{M}/L$. Fixed constant multiples of either saving only change the resulting absolute constant. $\square$

Below, j indexes a c -block and k its distance from $A = D$. Layer masses are $O(N\mathcal{L})$, so take $\mathcal{M} \asymp N\mathcal{L}^2$ and $L \asymp \mathcal{L}$. The block $j = 0$ is treated separately, never with $1/j$.

Theorem 8.2 (CONDITIONAL on (SEL)). *Assume (SEL) as stated in Section 1, uniformly for $\Gamma_0(4dq^2)$ and the even nebentypus characters modulo $2q$. Then, relative to the cited inputs listed in Section 1 (R114 repair),*

$$\sum_{p \leq N} f_I(p) \ll N \log^2 N.$$

Proof. We work with $N/2 < p \leq N$ and write $\mathcal{L} = \log N$. The reduction of Section 2, from [ET, Proposition 2.2 and Lemma 2.8], gives $f_I(p) \leq 2 \sum_c w_c(p)$. Fix small constants $0 < \eta \leq \eta_1$; all subsequent constants may depend on them.

1. *Removing the short-progression ranges.* Proposition 2.8 bounds the part $c > N^\eta$ by $O_\eta(N\mathcal{L}^2)$. For $c \leq N^\eta$, Lemma 2.6 disposes, all c at once, of tuples with

$$\min(4ab, 4acf, 4cdf) \leq N^{1-\eta_1}.$$

Indeed $bf = na + c$ and $f \leq 2n$ imply $b \geq a/2$, so $4ad$ and $4bd$ are already $\gg N^{1-\eta}$; no other short modulus is needed. The remaining tuples lie in $\mathcal{R}_{\text{bad}}(\eta_1)$ up to $O(1/\mathcal{L})$. Use nonnegative smooth cells of bounded overlap supported in $\mathcal{R}_{\text{bad}}(2\eta_1)$. Write

$$\begin{aligned} c &= N^\gamma, & a &\asymp A = N^\alpha, & d &\asymp D, & AD &\asymp N/c, \\ e &\asymp E = N^{\beta-\gamma}, & f &\asymp F = N^{1+\alpha-\beta}. \end{aligned} \tag{8.1}$$

Cells have side $O(1/\mathcal{L})$ in the exponent variables. The spectral weights may instead be chosen in $(f', a/f')$, $f' \in \{e, f\}$; these coordinates give the same bounded-overlap covering. Positive majorants allow us to drop the original size restrictions inside a cell. Proposition 2.3 gives, throughout the enlarged bad region,

$$e, f \gg N^{1/2-3\eta_1}. \tag{8.2}$$

For example $\beta \geq (1 - 2\eta_1)/2 + O(1/\mathcal{L})$, while $1 + \alpha - \beta \geq \max(1 - \alpha - \gamma - 2\eta_1, \alpha - 2\eta_1) + O(1/\mathcal{L})$.

2. *Bands and layers.* Let $c \asymp 2^j$, $0 \leq j \ll \eta\mathcal{L}$, and put

$$\delta = |2\alpha - 1 + \gamma|, \quad k = \lfloor \delta\mathcal{L} \rfloor.$$

A layer means fixed (j, k) and all its β -cells; there are only $O(1)$ choices of the a -block for each c -block and k . Fixed shifts in j, k and fixed changes of logarithm base are absorbed in constants. Remove bands of width $C_0/\mathcal{L}$ about $\beta = \alpha + \gamma$ and $\beta = 1$, and also the cells with $D \leq 64$. In each such cell one of e, f is $O_{C_0}(A)$ (for the last band use $ef = 4a^2d + 1$). Lemma 2.7(b), with a fixed enlarged constant

in $F \leq 4A$, bounds its mass by $O(2^j AD) = O(N)$. There are $O(\mathcal{L})$ band cells per c -block, so all bands cost $O(N\mathcal{L}^2)$ without a sieve. Layers $k \leq C_1 \log \mathcal{L}$ are postponed.

For reference, Brun–Titchmarsh on $4ad$ bounds a whole layer by

$$O(N\mathcal{L}/j), \quad j \geq 1, \quad (8.3)$$

whenever $k \leq \mathcal{L}/3$. For fixed (a, d, f) the c -block maps into an interval of length $4ad2^j$ in one residue class modulo $4ad$. Brun–Titchmarsh in the interval form $\pi(x+y; m, b) - \pi(x; m, b) \leq 2y/(\varphi(m) \log(y/m))$ [MV] (R114 repair: not [ET, (A.10)], which concerns initial segments), with $m = 4ad$ and $y = 4ad2^j$, gives $8ad2^j/(\varphi(4ad)j \log 2)$ for a reduced class. In a nonreduced class there is at most one prime, which is absorbed by the same upper bound. Now $k \leq \mathcal{L}/3$ implies $A, D \geq N^{1/4}$ for our fixed small η , and Lemma 2.7(a2), summed over the divisors f , proves (8.3). For $j = 0$ use its trivial analogue $O(N\mathcal{L})$.

3. *The cell estimates away from the bands.* All errors in the next paragraphs are *summed absolute errors* normalized by AD (by N after the c -sum), using Cauchy–Schwarz and Lemma 6.3 for Theorem 6.2. The bad region gives $A \geq N^{1/3-2\eta_1}$, hence $A \geq \mathcal{L}^2$ for Lemma 2.7(a1). All spectral boxes satisfy $Y/\lambda \asymp (A\sqrt{d})^{-1} \ll N^{-1/3+2\eta_1}$ and $L_* \ll \mathcal{L}$, so Proposition 5.1’s exact height condition holds for large N . We write F' for the scale of the selected divisor.

(i) $D \geq A$. Use the fixed- a sequences of Proposition 7.1. By (7.5) and (8.2) their normalized error, summed with sieve weights through $q \leq Q$, is

$$\ll \mathcal{L}^C (Q^2 N^{-\delta} + Q N^{-c_0}), \quad c_0 = \frac{1}{2} - 4\eta_1 > 0.$$

There is no restriction on β . The bad region gives $\alpha \geq 1/3 - 2\eta_1$, hence $\delta \leq 1/3 + 4\eta_1$ here. Thus $Q \leq N^{\delta/4}$ makes the second term an absolute power saving, for sufficiently small η_1 .

(ii) $D < A$ and $\beta < \alpha + \gamma$. Here $e \ll A$. If $k \geq 2j$ (allowing fixed shifts of the thresholds), use the e -cusp variant of Theorem 6.2, whose density is unchanged by the involution proved in §6. Thus $F' = E$ and $\lambda \asymp A/(qE)$, including $\lambda > 1$. Its summed error is

$$\ll \mathcal{L}^C Q^2 \left[N^{-\delta/2} (1 + A/E)^{1/2} + \frac{E^{1/2+\varepsilon}}{A} \right]. \quad (8.4)$$

Since $b \geq a/2$, we have $\beta \geq \alpha - O(1/\mathcal{L})$ and $(A/E)^{1/2} \ll N^{\gamma/2} \ll N^{\delta/4}$ in this range. The other term is $\ll A^{-1/2+2\varepsilon}$ and is $O(N^{-\delta/4})$ on the bad region, for a fixed sufficiently small ε . This gives $\mathcal{L}^C Q^2 N^{-\delta/4}$. Cells with $a/2 \leq b < a$ are included: they satisfy $e \leq 3a/c$. If $k < 2j$, use (8.3) instead; its hypotheses hold because $j \ll \eta\mathcal{L}$.

(iii) $D < A$ and $\alpha + \gamma < \beta < 1$. Off the bands, both $e, f \geq 8A$. Choose the smaller divisor, with scale $F' = \min(E, F) \ll A\sqrt{D}$, as cusp parameter. Then $\lambda \ll 1/q$, and Theorem 6.2 gives

$$\ll \mathcal{L}^C Q^2 \left[N^{-\delta/2} + N^\varepsilon D^{1/4} A^{-1/2} \right].$$

The second exponent is $(1 - \gamma - 3\alpha)/4 + \varepsilon$; its difference from $-\delta/4$ is $-\alpha/4 + \varepsilon$. It is therefore bounded by $N^{-\delta/4}$ with an absolute margin. We use the weaker common error $\mathcal{L}^C Q^2 N^{-\delta/4}$. (This corrects the source note [TTL, §8 (b3)], whose displayed $N^{-\delta/2}$ fails for $D < N^{4\varepsilon}$ or $\delta > 4/9$; the erratum R117 there does not affect the theorem, since step 4 already uses $\kappa = 1/4$ in (iii).)

(iv) $D < A$ and $1 < \beta \leq 1 + 2\eta_1$. This strip cannot be discarded: here $f < A$, even though e is large. Put $k' = \lfloor (\beta - 1)\mathcal{L} \rfloor$, which exceeds a fixed constant off the band. If $k' \leq k/2$, use the f -cusp, not the e -cusp. With $F' = F$ and $\lambda \asymp A/(qF)$, Theorem 6.2 gives

$$\ll \mathcal{L}^C Q^2 \left[N^{-\delta/2} (1 + A/F)^{1/2} + F^{1/2+\varepsilon}/A \right] \ll \mathcal{L}^C Q^2 N^{-\delta/4},$$

because $(A/F)^{1/2} \ll N^{(\beta-1)/2} \ll N^{\delta/4}$. The cusp term has an absolute margin. Crucially, the first term carries no N^ε loss at the degenerating boundary. If $k' > k/2$, use the cdf fibres of Lemma 2.6: $4cdf \asymp N^{2-\beta}$ and $\log(N/(4cdf)) \gg k'$. Brun–Titchmarsh costs $O(N/k')$ per cell. Indeed its reciprocal-totient sum is bounded by

$$\sum_{c \asymp 2^j} \frac{1}{\varphi(c)} \sum_{d \asymp D} \frac{1}{\varphi(d)} \sum_{f \asymp F} \frac{\rho_d(f)}{\varphi(f)} \ll 1$$

by Lemma 2.5(c). Comparing with this weighted sum bounds the nonreduced-class errors by $\sum_{c,d,f} \rho_d(f) \ll 2^j DF \asymp N \exp(-k') \ll N/k'$. On the sieve side the weighted cell mass is $O(N)$ by Lemma 2.7(b), since $F \leq A$. Thus in either treatment the cell costs $O(N/\max(k, k'))$ once the sieve saving below is established.

These cases cover every remaining cell: on $D \geq A$ use (i), and on $D < A$ the three intervals separated by $\alpha + \gamma$ and 1 give (ii)–(iv). Their boundaries and $A \asymp D$ were already set aside.

4. *Obtaining the prime saving from the cell errors.* For a sequence σ with fixed $s \in \{a, d\}$ and fixed c , choose

$$z = N^{\kappa\delta/8}, \quad Q = z^2, \quad \kappa = 1 \text{ in (i), } \quad \kappa = 1/4 \text{ in (ii)–(iv)}.$$

Sieve only by primes $\ell > \ell_0 \geq 3$ with $\ell \nmid 2cs$. Thus $(q, 2ca) = 1$ in Proposition 7.1, and $(q, 2cd) = 1$ in Theorem 6.2; in particular $\kappa_c(q) = 1$. Both densities satisfy

$$\frac{1}{\ell} - \frac{2}{\ell^2} \leq g_\sigma(\ell) \leq \frac{1}{\ell} < 1.$$

Selberg’s upper-bound sieve [HR, Theorem 4.1 and Lemma 4.1] for nonnegative weighted sequences therefore gives

$$\#\{n \in \sigma : n \text{ prime, } n > N/2\} \leq \frac{X_\sigma}{G_\sigma(z)} + \sum_{\substack{q \leq Q \\ q|P_\sigma(z)}} 3^{\omega(q)} |r_\sigma(q)|, \quad (8.5)$$

$$G_\sigma(z) \gg \frac{\varphi(2cs)}{2cs} \log z.$$

Here counts are weighted if the cell is smooth; $z < N/2$. The lower bound on the local density, not merely its upper bound, is used in the estimate for G_σ .

The model mass X_σ need not be compared from below to the actual mass. At $q = 1$, Theorem 6.2 or Proposition 7.1 gives

$$X_\sigma \leq \#\sigma + |r_\sigma(1)|.$$

Also $2cs/\varphi(2cs) \leq 2g(c)g(s)$. Lemma 2.7(a1) and $\sum_{c \asymp 2^j} g(c) \ll 2^j$ bound the weighted mass of a layer in (i) or (iii) by $O(N\mathcal{L})$. Lemma 2.7(b) bounds each cell in (ii) or (iv) by $O(N)$; in particular their layer masses are $O(N\mathcal{L})$. The $q = 1$ errors introduced in this comparison have only the extra factor $\max g(c)g(s) \ll (\log \mathcal{L})^2$. Since $Q^2 = N^{\kappa\delta/2}$, the summed errors are $\ll \mathcal{L}^C N^{-\kappa\delta/2}$ times the relevant mass scale; the additional term in (i) has an absolute power saving. All these errors are

$O(1/k)$ times that scale once $k \geq C_1 \log \mathcal{L}$, with C_1 fixed sufficiently large. The main term in (8.5) has the same saving, since $\log z \asymp k$. This proves all the sieve savings asserted in step 3.

5. *Summing all cells.* For $1 \leq j \ll \eta \mathcal{L}$, the layers $k \leq C_1 \log \mathcal{L}$ cost, by (8.3),

$$\ll N \mathcal{L} \log \mathcal{L} \sum_{j \ll \eta \mathcal{L}} \frac{1}{j} \ll N \mathcal{L} (\log \mathcal{L})^2 \ll N \mathcal{L}^2.$$

For $j = 0$ these layers cost $O(N \mathcal{L} \log \mathcal{L})$ trivially. In (i), (iii), and (ii) with $k \geq 2j$, a layer costs

$$\ll N \mathcal{L} \min(1/j, C/k) \quad (k \leq \mathcal{L}/3).$$

One takes the better bound for the whole portion of the layer in question, not an unweighted average of the two methods. Lemma 8.1 sums these costs to $O(N \mathcal{L}^2)$. For $k > \mathcal{L}/3$ the sieve alone suffices: $\sum_j \sum_{k > \mathcal{L}/3} N \mathcal{L}/k \ll N \mathcal{L}^2$. Case (ii) with $k < 2j$ costs $\sum_j \sum_{k < 2j} O(N \mathcal{L}/j) = O(N \mathcal{L}^2)$. In (iv) the per-cell bound gives, for each c -block,

$$\sum_{k, k' \ll \mathcal{L}} \frac{CN}{\max(k, k')} \ll N \sum_{h \ll \mathcal{L}} \frac{2h-1}{h} \ll N \mathcal{L}.$$

Summing over j gives $O(N \mathcal{L}^2)$ again. For $j = 0$ outside the innermost layers the $1/k$ bound costs at most $O(N \mathcal{L} \log \mathcal{L})$, and the preceding two-dimensional sum still applies to (iv). Together with the bands and step 1, this proves the bound on $(N/2, N]$. Summing dyadically in N proves the theorem. $\square$

9. THE EXCEPTIONAL SPECTRUM ON AVERAGE OVER THE LEVEL

This section removes (SEL) from Theorem 8.2; it is a write-up of [TTL2]. With the published, unquantified ε -constant of the cited input it gives Theorem 9.9(i), and with the effective constant of Appendix A it gives Theorem 9.9(ii). Throughout, $\mathcal{L} \geq 2$ is as in Proposition 5.1 (we may take $\mathcal{L} = \log N$ in the application), and an exceptional form of $(\Gamma_0(M), \chi)$ is an orthonormal cusp form u_j with $t_j = i\sigma_j$, $0 < \sigma_j < 1/2$; Selberg's bound [DI, Theorem 4] gives $\sigma_j \leq 1/4$. (The value $t_j = 0$, $\lambda_j = 1/4$, is not exceptional.) We use two facts about the exceptional spectrum: the cited bound for $a_n = 1$ below, and partial summation, which reduces the smooth unfolded coefficients of a box to $a_n = 1$. The decisive point is that the assembly of Section 8 already sums the per- d errors over $d \asymp D$, and $d \mapsto M = 4dq^2$ is an average over levels.

The cited input. For $q_0 = 1$ this is [DI, Theorem 7, (1.41), p. 233], quoted verbatim (checked against the scan, with $\Sigma^{(q)}$ defined on the same page: “The sum $\Sigma^{(q)}$ is extended to the λ_j ’s belonging to the exceptional eigenvalues of the Hecke group $\Gamma_0(q)$ ”):

Theorem 7. *Let $Q, N, X \geq 1$ and ε be any positive constant. Then we have*

$$\sum_{q \leq \mathfrak{Q}_j\text{-except}} \sum^{(q)} X^{4i\kappa_j} \left| \sum_{n \leq N} \rho_{j\infty}(n) \right|^2 \ll (QN)^\varepsilon (Q + N + \sqrt{N}X)N,$$

the constant implied in $\ll$ depending on ε alone.

Here $i\kappa_j = \sigma_j$, so $X^{4i\kappa_j} = Y^{2\sigma_j}$ with $Y = X^2$, and $\sqrt{N}X = \sqrt{NY}$; this is the normalisation of the proof, [DI, (8.17)–(8.19), pp. 276–278]. (Below (8.18) DI print $Y_1 = \sqrt{Q+N}$, which should read $Y_1 = Q+N$; this misprint does not affect the theorem.) The nebentypus version is [Dr, Lemma 4.10, arXiv p. 16], quoted verbatim:

Lemma 4.10. *Assume that the situation is as in Lemma 4.9. Assume moreover that $(a_n)_{N < n \leq 2N}$ is the characteristic sequence of an interval of integers. Then*

$$\sum_{\substack{q \leq Q \\ q_0 | q}} \mathcal{E}_{q,\infty}(Y, (a_n)) \ll_\varepsilon (QN)^\varepsilon (Qq_0^{-1} + N + (NY)^{1/2})N.$$

In Lemma 4.9 the multiplier χ has modulus $q_0 \geq 1$, $Y \geq 1$, $Q \geq q_0$, and the setting of [Dr, Proposition 4.7] has $N \geq 1/2$;

$$\mathcal{E}_{q,\mathfrak{a}}(Y, (a_n)) = \sum_{\substack{f \in \mathcal{B}(q, \chi) \\ t_f \in i\mathbb{R}}} Y^{2|t_f|} \left| \sum_{N < n \leq 2N} a_n n^{1/2} \rho_{f\mathfrak{a}}(n) \right|^2$$

[Dr, §4.2.3], where $\mathcal{B}(q, \chi)$ is an orthonormal basis of Maass cusp forms for $\Gamma_0(q)$ with multiplier χ . As recorded in Section 5, $n^{1/2} \rho^{\text{Dr}}(n) = \rho^{\text{DI}}(n)/2$, and $Y^{2|t_f|} = Y^{2\sigma_j}$. Drappeau proves Lemma 4.10 by transposing DI’s proof (pp. 274–277); we use his statement as a cited input, uniformly in q_0 , as he does in [Dr, §4.3.1]. (We read his $\ll_\varepsilon$ as uniform in q_0 ; he remarks that q_0 enters the error terms only with negative powers.) Every summand is nonnegative, so the bound also holds for any subset of the levels $q \leq Q$, $q_0 \mid q$.

Corollary 9.1 (prefix sums; PROVED relative to [DI, Dr]). *For each $\varepsilon \in (0, 1/4]$ there is C_ε such that the following holds. Let $q_0 \geq 1$, χ a character modulo q_0 , $M_0 \geq q_0$, $\mathcal{M}$ a set of levels $M \leq M_0$ with $q_0 \mid M$, $t \geq 1$ and $Y \geq 1$. Put $S_j(t) = \sum_{1 \leq n \leq t} \rho_{j\infty}(n)$. Then*

$$\sum_{M \in \mathcal{M}} \sum_{j \text{ exc } (M, \chi)} Y^{2\sigma_j} |S_j(t)|^2 \leq C_\varepsilon (M_0 t)^\varepsilon \log^2(2t) (M_0 + t + \sqrt{tY}) t.$$

Proof. Cover $[1, t]$ by the $K \leq 2 + \log_2 t$ (R118 repair) intervals $I_i = (2^{i-1}, \min(2^i, t)]$, $i \geq 0$; each is an interval of integers inside $(N_i, 2N_i]$, $N_i = 2^{i-1} \in [1/2, t]$. Cauchy–Schwarz gives $|S_j(t)|^2 \leq K \sum_i |\sum_{n \in I_i} \rho_j(n)|^2$. Apply Lemma 4.10 to each i with $Q = M_0$, positivity, and $Q/q_0 + N_i + \sqrt{N_i Y} \leq M_0 + t + \sqrt{tY}$, $N_i \leq t$. Since $K^2 \ll \log^2(2t)$, the claim follows. (A factor $\log(2t)$ would suffice.) $\square$

For $q_0 = 1$ only DI’s Theorem 7 is used. We write (DI7_ε) for Corollary 9.1 with the constant C_ε .

Lemma 9.2 (partial summation; PROVED). *Let $c : [1, \infty) \rightarrow \mathbb{C}$ be C^1 with $|c(t)| + |c'(t)| \ll_B t^{-B}$ for every B , and let $\Psi, \Psi' \geq 0$ be measurable with $|c'(t)| \leq \Psi(t)\Psi'(t)$ and $\int_1^\infty \Psi < \infty$. Then*

$$\left| \sum_{n \geq 1} c(n) \rho_j(n) \right|^2 \leq \left(\int_1^\infty \Psi \right) \int_1^\infty \Psi(t) \Psi'(t)^2 |S_j(t)|^2 dt.$$

Proof. Abel summation gives $\sum_{n \geq 1} c(n) \rho_j(n) = -\int_1^\infty S_j(t) c'(t) dt$; the boundary term vanishes because $S_j(t) \ll_j t^{3/2}$ and c decays rapidly. Apply Cauchy–Schwarz with the measure $\Psi(t) dt$. $\square$

The exceptional part of the box variance, averaged over d . Keep the notation of Proposition 5.1 and its proof, and write $\mathcal{W}^*(v) = \sup_{0 \leq \sigma \leq 1} |\mathcal{W}(\sigma + iv)|$ (R118 repair: $\Re s_v = \sigma_j + 1/\mathcal{L}$ may exceed $1/2$), so that $\mathcal{W}^*(v) \ll_B (1 + |v|)^{-B}$.

Lemma 9.3 (exceptional coefficients on a shifted line; PROVED). *Let u_j be exceptional for $(\Gamma_0(M), \chi)$, put $s_v = \sigma_j + 1/\mathcal{L} + iv$ and $\tilde{B}_j(s) = \sum_{n \neq 0} \overline{\rho_j(n)} \lambda \hat{\phi}(\lambda n) (\pi |n| Y)^{-s}$. Then*

$$|\langle P_\chi, u_j \rangle|^2 \ll \mathcal{L}^4 Y^{-1} \int_{\mathbb{R}} \mathcal{W}^*(v) |\tilde{B}_j(s_v)|^2 dv.$$

Proof. For $t_j = i\sigma_j$ the Mellin formula for K_{it_j} , hence (5.6), holds on every line $\Re s = \sigma > \sigma_j$; take $\sigma = \sigma_j + 1/\mathcal{L}$. Both arguments of $G_{t_j}(s_v) = \Gamma((s_v - \sigma_j)/2) \Gamma((s_v + \sigma_j)/2)$ have real part in $[1/(2\mathcal{L}), 1/2]$, so $|G_{t_j}(s_v)| \ll \mathcal{L}^2 e^{-\pi|v|/2}$ (use $|\Gamma(x + iy)| \leq \Gamma(x) \ll 1/x$ for $|y| \leq 1$ and Stirling otherwise). Apply Cauchy–Schwarz in v with the measure $|G_{t_j} \mathcal{W}| dv$. No contour is shifted, so the poles of Γ at $\pm \sigma_j$ never enter. $\square$

Proposition 9.4 (averaged exceptional variance; PROVED relative to [DI, Dr]). *Let q be squarefree, $\lambda > 0$, $D \geq 1$, $0 < Y_0 \leq 1$, and for each $d \in (D, 2D]$ with $(q, 2d) = 1$ let $\psi_d(u) = \phi(x/\lambda) W(y/Y_d)$ with $Y_d \in [Y_0/\sqrt{2}, Y_0]$, on $\Gamma_{M,q}$, $M = 4dq^2$. Let*

$$\mathcal{E}_d = \frac{1}{h_q} \sum_{\chi \text{ even mod } q} \sum_{j \text{ exc } (M, \chi)} |\langle P_\chi, u_j \rangle|^2$$

be the exceptional part of (5.5) for $\psi = \psi_d$. Put $M_0 = 8Dq^2$, $\lambda_- = \min(\lambda, 1)$, and let $\mathcal{L} \geq \log(2 + (\lambda_- Y_0)^{-1})$. Then for every $\varepsilon \in (0, 1/4]$,

$$\sum_d \mathcal{E}_d \ll \mathcal{L}^C C_\varepsilon (M_0/\lambda_-)^\varepsilon Y_0^{-1} [\lambda_- M_0 + 1 + \lambda_- Y_0^{-1/2}],$$

where C and the implied constant depend only on ϕ, W .

Proof. Write $\tilde{B}_j(s) = \sum_{n \geq 1} \overline{\rho_j(n)} c_+(n) + \sum_{n \geq 1} \overline{\rho_j(-n)} c_-(n)$, with $c_\pm(t) = \lambda \hat{\phi}(\pm \lambda t) (\pi t Y_d)^{-s}$. The map $u(z) \mapsto u(-\bar{z})$ preserves $(\Gamma_0(M), \chi)$ (conjugation by $\text{diag}(1, -1)$ fixes the lower-right entry), the exceptional spectrum and σ_j , and exchanges $\rho_j(-n)$ and $\rho_j(n)$; so it suffices to treat $n > 0$. Put

$$w(t) = \max(1, (\pi t Y_0)^{-1}), \quad \Phi(t) = \max_{\pm} (\lambda^2 |\hat{\phi}'(\pm \lambda t)| + \lambda |\hat{\phi}(\pm \lambda t)|/t).$$

For $s = s_v$ we have $|(\pi t Y_d)^{-s}| \leq e^2 w(t)^{\sigma_j}$, using $Y_d \geq Y_0/\sqrt{2}$, $\sigma_j \leq 1/4$ and $Y_0^{-1/\mathcal{L}} \leq e$; hence $|c'_\pm(t)| \leq e^2 (2 + |v|) \Phi(t) w(t)^{\sigma_j}$. Lemma 9.2 with $\Psi = \Phi$ and $\Psi' = e^2 (2 + |v|) w^{\sigma_j}$ gives

$$\left| \sum_n \overline{\rho_j(n)} c_+(n) \right|^2 \leq e^4 (2 + |v|)^2 \left(\int_1^\infty \Phi \right) \int_1^\infty \Phi(t) w(t)^{2\sigma_j} |S_j(t)|^2 dt.$$

The weight $w(t)^{2\sigma_j}$ depends on t ; it is the size of $(tY)^{-2\sigma_j}$ when tY is small. Insert this into Lemma 9.3, use $Y_d^{-1} \leq \sqrt{2} Y_0^{-1}$, and sum over j, d and χ . For each t and each χ apply (DI7 $_\varepsilon$) with $q_0 = q$ (the modulus of χ), the levels $\{4dq^2 : d \in (D, 2D]\}$

(all $\leq M_0$ and divisible by q), and $Y = w(t) \geq 1$. Since $tw(t) = \max(t, (\pi Y_0)^{-1})$, $\sqrt{tw(t)} \leq \sqrt{t} + Y_0^{-1/2}$. The average over the h_q characters costs nothing. Hence

$$\sum_d \mathcal{E}_d \ll \mathcal{L}^4 Y_0^{-1} \left(\int_1^\infty \Phi \right) \int_1^\infty \Phi(t) C_\varepsilon(M_0 t)^\varepsilon \log^2(2t) (M_0 + t + Y_0^{-1/2}) t dt,$$

where $\int \mathcal{W}^*(v)(2 + |v|)^2 dv \ll 1$. As $\hat{\phi}$ is a Schwartz function, $\int_1^\infty \Phi \ll \mathcal{L} \lambda_-$ and, uniformly for $0 \leq a \leq 2$ and $\varepsilon \leq 1/4$, $\int_1^\infty \Phi(t) t^{1+a+\varepsilon} \log^2(2t) dt \ll \mathcal{L}^2 \lambda_-^{-a-\varepsilon}$ (substitute $u = \lambda t$ if $\lambda \leq 1$; for $\lambda > 1$ everything is $\ll \lambda^{-B}$). Use $a = 0$ for the terms M_0 and $Y_0^{-1/2}$, and $a = 1$ for the term t . $\square$

Remark 9.5 (why the weight must depend on n). With the cruder constant weight $Y_0^{-2\sigma_j} \geq w(t)^{2\sigma_j}$ the third term would be $\lambda_-^{1/2} Y_0^{-3/2}$ instead of $\lambda_- Y_0^{-3/2}$. This is larger by $\lambda_-^{-1/2}$, which is fatal in case (iii) of Theorem 8.2, where λ can be as small as $D^{-1/2}$. The structure $a_n = 1$ is used exactly here: partial summation absorbs the whole n -range at once. The level average in [DI, Theorem 6], for general a_n , has NX in place of $X\sqrt{N}$ and does not suffice; [DI, Theorem 5] has no level average.

Per- d counts averaged over d , without (SEL). In the proof of Proposition 5.1, (SEL) was used only to discard (a) exceptional cusp forms and (b) a nonconstant residual spectrum. Point (b) needs no hypothesis: $\Gamma_{M,q}$ is a congruence subgroup, and the Eisenstein series of a congruence subgroup have no pole in $1/2 < \Re s < 1$, so its residual spectrum consists of the constants [Iw, §11], [Hux] (a cited standard fact; the locators were not re-verified). Every other step of that proof, and the large sieve (5.2), is applied to the cusp forms with $t_j \in \mathbb{R}$ and to the continuous spectrum; restricting a large sieve to a subset of the spectrum only decreases its left side. Hence, *unconditionally*,

$$\begin{aligned} \|(1 - \Delta)(P_\psi - \langle P_\psi \rangle)\|_2^2 &\ll V^{\text{gen}} + \mathcal{E}^{(1)} + \mathcal{E}^{(2)}, \\ V^{\text{gen}} &\ll \mathcal{L}^C \left\{ \frac{\lambda}{Y} \left(1 + \frac{q^{1/2}}{M} (\mathcal{L}/\lambda)^{1+\varepsilon_1} \right) + \frac{\lambda^2}{Y} \right\}, \end{aligned} \quad (9.1)$$

with the fixed exponent $\varepsilon_1 = 1/100$ (so that its constant is absolute), where $\mathcal{E}^{(1)}, \mathcal{E}^{(2)}$ are the exceptional parts of Proposition 9.4 for the two boxes into which $(1 - \Delta)\psi$ decomposes at the start of the proof of Proposition 5.1. (R118 repair: the second box carries the d -dependent factor $(Y_d/\lambda)^2 \leq 1$; dropping it only enlarges $\mathcal{E}^{(2)}$, so Proposition 9.4 applies with fixed weights.)

Theorem 9.6 (averaged per- d count; PROVED relative to [DI, Dr]). *Let $q \leq N^{1/50}$ be squarefree (R118 repair: the sieve uses $q \leq N^{\delta/64} \leq N^{1/64}$), and $(q, 2c) = 1$. For $d \in (D, 2D]$, $(q, 2d) = 1$, let ψ_d be the boxes of Theorem 6.2 (or of its e -cusp variant) with a common $\lambda \asymp A/(qF')$ and $Y_d = 1/(2qF'\sqrt{d})$, where $F' \geq 1$ is the scale of the selected cusp divisor, $F' \ll A\sqrt{D}$ (R118 repair: only this order of magnitude is used), $D \leq N$, and $Y_d \leq \lambda \mathcal{L}^{-3}$. Write $E_d(q) = \sum_{Q \in \mathcal{F}_d^I, q|n(Q)} \psi_d(u_Q) - g_{c,d}(q) \mathfrak{M}_d$. Then for every $\varepsilon \in (0, 1/4]$,*

$$\begin{aligned} \sum_d |E_d(q)| &\ll \mathcal{L}^C C_\varepsilon^{1/2} N^{2\varepsilon} AD \left[q^2 \left(\frac{D}{A} \right)^{1/2} \left(1 + \frac{A}{qF'} \right)^{1/2} \right. \\ &\quad \left. + \frac{q^{3/2} N^{\varepsilon_1/2} F'^{1/2}}{A} + \frac{q^{5/4} F'^{1/4} D^{1/8}}{A^{1/2}} \right]. \end{aligned}$$

Proof. The proof of Theorem 6.2 gives $|E_d(q)| \ll q(\#\Lambda_d(1))^{1/2} V_d^{1/2}$, with V_d the left side of (9.1) for ψ_d ; this step (Sobolev duality and Lemma 6.1) does not use (SEL). Cauchy–Schwarz over d and Lemma 6.3 give $\sum_d |E_d(q)| \ll \mathcal{L}^C q D^{3/4} (\sum_d V_d)^{1/2}$. Put $Y_0 = 1/(2qF'\sqrt{D})$, so $Y_d \in [Y_0/\sqrt{2}, Y_0]$ and $\lambda/Y_0 \asymp A\sqrt{D}$. By (9.1), summed over the $\leq D$ values of d with $M \geq 4Dq^2$,

$$\sum_d V_d^{\text{gen}} \ll \mathcal{L}^C [DA\sqrt{D}(1+\lambda) + N^{\varepsilon_1} q^{-3/2} Y_0^{-1}].$$

Proposition 9.4 (applied with $\mathcal{L} \asymp \log N$ large enough that $\mathcal{L} \geq \log(2 + (\lambda_- Y_0)^{-1})$, which only changes $\mathcal{L}^C$; R118 repair), with $M_0 = 8Dq^2$, $\lambda_- M_0 \leq 8q^2 D\lambda$ and $M_0/\lambda_- \leq N^3$ (as $1/\lambda_- \ll \max(1, qF'/A) \leq N^{2/3}$), gives $\sum_d \mathcal{E}_d \ll \mathcal{L}^C C_\varepsilon N^{3\varepsilon} [q^2 DA\sqrt{D} + Y_0^{-1} + \lambda_- Y_0^{-3/2}]$. With $Y_0^{-1} \asymp qF'\sqrt{D}$ and $\lambda_- Y_0^{-3/2} \ll A\sqrt{D}(qF'\sqrt{D})^{1/2}$, the three resulting terms, multiplied by $qD^{3/4}$ and divided by AD , are the three terms of the claim. $\square$

Under (SEL) the exceptional part vanishes and the first term has q in place of q^2 ; the extra q and the third term are the price of the exceptional spectrum.

Corollary 9.7 (the cases (ii)–(iv) without (SEL)). *In the cells of cases (ii) (with $k \geq 2j$), (iii) and (iv) (with $k' \leq k/2$) of the proof of Theorem 8.2, with the cusp parameter chosen there and $\delta = 2\alpha - 1 + \gamma > 0$, the sum in Theorem 9.6 is $\ll \mathcal{L}^C C_\varepsilon^{1/2} N^{2\varepsilon} q^2 N^{-\delta/4} AD$. No upper bound on δ is needed.*

Proof. On the side $D < A$ we have $D/A \asymp N^{-\delta}$. First term: in (ii) and (iv), $(1+A/(qF'))^{1/2} \ll N^{\delta/4}$ as shown in step 3 of the proof of Theorem 8.2; in (iii) $\lambda \leq 1$. In all three cases $F' \ll A\sqrt{D}$: in (ii) $E \ll A$, in (iii) $\min(e, f) \leq (ef)^{1/2} \leq 3a\sqrt{d}$, in (iv) $F < A$. Hence the third term is $\ll q^{5/4} (A\sqrt{D})^{1/4} D^{1/8} A^{-1/2} \ll q^{5/4} (D/A)^{1/4}$ (R118 repair: $q^{5/4}$ restored), and the second is $\ll q^{3/2} N^{\varepsilon_1/2} A^{-1/4} (D/A)^{1/4} \leq q^2 (D/A)^{1/4}$, because $A \geq N^{1/3-2\eta_1}$ on the bad region. $\square$

The Type I sum.

Theorem 9.8 ((EFF); PROVED relative to (B1)–(B5), Appendix A). *There is an absolute A_0 such that (DI7 $_\varepsilon$) holds with $C_\varepsilon \leq \exp(\exp(A_0/\varepsilon))$ for all $\varepsilon \in (0, 1/4]$, for DI’s Theorem 7 and for Drappeau’s Lemma 4.10, uniformly in q_0 and in the even character χ modulo q_0 .*

This is Theorem A.1. The published proofs [DI, Dr] state $\ll_\varepsilon$ without specifying the dependence on ε ; Appendix A makes them effective (black boxes (B1)–(B5): exact trace formulae, Weil’s bound and a twisted Weil bound, Selberg’s $3/16$, and $\lambda_1(\text{SL}_2(\mathbb{Z})) > 1/4$). Only even χ occur below. In earlier versions of this note (EFF) was a hypothesis, supported only by an audit (Remark 9.11).

Theorem 9.9. *Let $\mathcal{L} = \log N$.*

- (i) PROVED relative to the cited inputs of Section 1, [DI, Theorem 7] and [Dr, Lemma 4.10]; no spectral hypothesis. *There is an absolute C such that for every $\varepsilon_0 \in (0, 1/4]$*

$$\sum_{p \leq N} f_I(p) \leq C\varepsilon_0 N \mathcal{L}^2 \log \mathcal{L} + O_{\varepsilon_0}(N \mathcal{L}^2).$$

In particular $\sum_{p \leq N} f_I(p) = o(N \log^2 N \log \log N)$.

- (ii) PROVED relative to the cited inputs of Section 1 and to Theorem 9.8 (i.e. to (B1)–(B5) and Appendix A): $\sum_{p \leq N} f_I(p) \ll N \log^2 N$.
- (iii) (quantified form) If $C_\varepsilon \leq G(1/\varepsilon)$ with G nondecreasing, then $\sum_{N/2 < p \leq N} f_I(p) \ll N \mathcal{L}^2(1 + w \log \mathcal{L})$ for every $w \in [\mathcal{L}^{-1/2}, 1/4]$ with $\log G(64/w) \leq w \mathcal{L}/16$, the implied constant independent of w and G .

Proof. Work on $(N/2, N]$ as in the proof of Theorem 8.2. That proof uses (SEL) only through Theorem 6.2 in cases (ii) (with $k \geq 2j$), (iii) and (iv) (with $k' \leq k/2$), all on the side $D < A$, and there only through $\sum_{d \asymp D} |E_d(q)|$ (step 3: Cauchy–Schwarz and Lemma 6.3). Steps 1 and 2, case (i), the Brun–Titchmarsh alternatives and the bands are unconditional. Replace that use by Corollary 9.7, and in step 4 take $\kappa = 1/16$ in (ii)–(iv), so $z = N^{\delta/128}$, $Q = z^2 = N^{\delta/64} \leq N^{1/64}$ (as $\delta \leq 1$, so Theorem 9.6 applies to every $q \leq Q$; R118 repair), and the main-term saving is still $1/G_\sigma(z) \ll g(c)g(s)/k$. By $\sum_{q \leq Q} 3^{\omega(q)} q^2 \ll \mathcal{L}^C Q^3$, the sieve remainder and the $|r_\sigma(1)|$ part of the main term are, relative to the mass,

$$\ll \mathcal{L}^C C_\varepsilon^{1/2} N^{2\varepsilon} N^{-\delta/4+3\delta/64}.$$

Bookkeeping in ε . Fix a threshold $w \in (0, 1/4]$ and put $\varepsilon = w/64$. For $\delta \geq w$, $N^{2\varepsilon} \leq N^{\delta/32}$ and the display is $\leq \mathcal{L}^C C_\varepsilon^{1/2} N^{-11\delta/64}$. Call a layer *good* if $\delta \geq w$ and $\mathcal{L}^C C_\varepsilon^{1/2} N^{-11\delta/64} \leq 1/(\delta \mathcal{L})$. On good layers steps 4 and 5 of the proof of Theorem 8.2 apply verbatim (saving C/k , summed by Lemma 8.1).

The residual strip. We show below that every other layer of the side $D < A$ has $\delta < w \leq 1/4$, hence $k \leq \mathcal{L}/3$; there the Brun–Titchmarsh bound (8.3) on $4ad$ is applied to the whole layer. It costs $O(N\mathcal{L}/j)$ for each c -block $j \geq 1$ and $O(N\mathcal{L})$ for $j = 0$, i.e. $O(N\mathcal{L} \log \mathcal{L})$ per layer after summing over $j \ll \eta \mathcal{L}$. At most $w\mathcal{L} + 1$ layers have $\delta < w$, so the strip costs $\ll (w\mathcal{L} + 1)N\mathcal{L} \log \mathcal{L}$.

(i) Take $w = \varepsilon_0$. Then $C_\varepsilon = C_{\varepsilon_0/64}$ is a constant and $N^{-11\delta/64} \leq N^{-\varepsilon_0/6}$, so every layer with $\delta \geq \varepsilon_0$ is good once $N \geq N_0(\varepsilon_0)$. The strip costs $\leq C_{\varepsilon_0} N \mathcal{L}^2 \log \mathcal{L} + O(N\mathcal{L} \log \mathcal{L})$; the rest is $O_{\varepsilon_0}(N\mathcal{L}^2)$ by the proof of Theorem 8.2. Sum dyadically in N .

(ii) Take $w = w_N = 128A_0/\log \mathcal{L}$, which is $\leq 1/4$ for N large (this affects only the range of N , not the constant). Then $\log C_\varepsilon \leq \exp(64A_0/w) = \mathcal{L}^{1/2}$, and for $\delta \geq w_N$, $\frac{11}{64}\delta \mathcal{L} \geq 22A_0\mathcal{L}/\log \mathcal{L} \geq \frac{1}{2}\mathcal{L}^{1/2} + (C+1)\log \mathcal{L}$ for $\mathcal{L} \geq \mathcal{L}_0(A_0)$; so every layer with $\delta \geq w_N$ is good. The strip costs $\ll (w_N \mathcal{L} + 1)N\mathcal{L} \log \mathcal{L} \ll A_0 N \mathcal{L}^2$.

(iii) As in (ii): for $\delta \geq w$, $\frac{1}{2}\log C_\varepsilon \leq \frac{1}{2}\log G(64/w) \leq \delta \mathcal{L}/32$, and $\frac{11}{64}\delta \mathcal{L} - \frac{2}{64}\delta \mathcal{L} = \frac{9}{64}\delta \mathcal{L} \geq (C+1)\log \mathcal{L}$ as $\delta \geq \mathcal{L}^{-1/2}$, for $\mathcal{L}$ large. The strip costs $\ll (w\mathcal{L} + 1)N\mathcal{L} \log \mathcal{L}$. $\square$

Remark 9.10. The side $D \geq A$ never used (SEL): it is covered by the per- a Weil count (Proposition 7.1). The “strip” of the earlier unconditional discussion is therefore exactly the set of non-good layers, contained in $0 < \delta < w$; its cost $wN\mathcal{L}^2 \log \mathcal{L}$ is $O(N\mathcal{L}^2)$ if and only if $w \ll 1/\log \mathcal{L}$, which is what (EFF) (Theorem 9.8) provides. Nothing about the location of exceptional eigenvalues remains: Kim–Sarnak is not used, and only Selberg’s $\sigma_j \leq 1/4$ enters, inside DI’s proof.

Remark 9.11 (history: the effectivity audit, superseded by Appendix A). Before Appendix A was written, an internal audit of DI pp. 228–236, 242–245, 253–278 and of [Dr, §4.2.3] locates the ε -losses in divisor bounds $\tau(k) \ll k^\varepsilon$, logarithmic factors, an integration by parts of order about $2/\varepsilon$ [DI, (1.27), p. 257], and the inductions

on Q [DI, (8.12), (8.19)]. No ineffective input was found: DI's Theorem 14 uses Theorem 9 only for the full modular group, which has no exceptional spectrum, and Selberg's $3/16$ is explicit. With DI's unspecified C^∞ cutoff chosen of Gevrey class 2, the bookkeeping suggests $C_\varepsilon \leq \exp(\exp(A/\varepsilon))$. For Drappeau's nebenspectrum version the same two inductions appear; his twisted trace formulae were not audited, so (EFF) there was only very plausible. Appendix A turns this assessment into a proof; on the way the twisted trace formulae were checked, and the nebenspectrum induction turned out to need an extra small- C branch (Proposition A.3, case (C2)). Independently of the appendix, any published form of the two estimates with $(QN)^\varepsilon$ replaced by $\exp(o(\log(QN)/\log \log(QN)))$ would suffice via Theorem 9.9(iii).

Remark 9.12 (general numerators). The internal note [MN4] carries the conditional argument over to m/p , with (SEL) replaced by its analogue (SEL _{m}) for the levels $\Gamma_0(mdq^2)$, and obtains $\sum_{N/2 < p \leq N} f_{I,m}(p) \ll N(\mathcal{L}^2 + \mathcal{L} \log^2 m)/m + Nm^{-0.35}/\mathcal{L}$ uniformly for $4 \leq m \leq \mathcal{L}^5$; with [MN2, Theorem U] this places the density transition for m/p at $\log N \asymp m^{1/3}$, with matching orders on both sides. That note predates the present section and is conditional on (SEL _{m}). Whether the level average used here transfers uniformly in m (levels mdq^2 , sieve omitting the primes dividing m) has not been checked.

10. SINGLE-LEVEL EXCEPTIONAL ESTIMATES

The separation argument, Sobolev duality, the local-density computation, the per- a count, and the arithmetic mass estimates are unconditional. In Theorem 8.2 the use of (SEL) is confined to the exceptional spectrum in Proposition 5.1. This section records why estimates *at a single level* do not suffice; it explains why the level average of Section 9 is needed, and is not used in any proof.

10.1. The exceptional-spectrum obstruction. An eigenvalue $\lambda_j = 1/4 - \sigma_j^2$, $0 < \sigma_j < 1/2$, introduces in the unfolded coefficient the factor $(nY)^{-\sigma_j}$, since $K_{\sigma_j}(2\pi ny)$ grows like a constant times $(ny)^{-\sigma_j}$ when ny is small. The unconditional Kim–Sarnak bound [KS, Appendix 2] is $\sigma_j \leq 7/64$, including the nebenspectrum forms at issue here. A direct use of this bound permits the loss

$$Y^{-7/64} \asymp (qF'\sqrt{d})^{7/64} \tag{10.1}$$

in the spectral error. On the $D < A$ side this must be compared with $(D/A)^{1/2} = N^{-\delta/2}$, $\delta = 2\alpha - 1 + \gamma > 0$. In the worst cells $qF'\sqrt{d}$ has size about N , so the exponent comparison leaves a strip

$$0 < 2\alpha - 1 + \gamma \lesssim 7/32.$$

Even the smallest cusp parameters at the boundary, with $F' \gg N^{1/2-3\eta_1}$ and D near $N^{1/2}$, give a loss of size $N^{21/256-O(\eta_1)}$, corresponding to the comparison $\delta \approx 21/128 - O(\eta_1)$. These numbers describe the obstruction to the estimates, not a sharp geometric boundary or a lower bound for an actual spectral contribution. In particular a fixed positive width, rather than a single line, remains untreated.

Assessment 10.1 (partial exceptional-spectrum estimates). Deshouillers–Iwaniec's exceptional large sieve [DI, Theorem 5] improves on a pointwise insertion of (10.1), but does not supply the needed uniform saving at $A \asymp D$. On a Fourier block

$n \asymp N_0$ its parameter must be

$$X = \frac{1}{N_0 Y},$$

not $1/Y$, because the squared exceptional weight is $(nY)^{-2\sigma_j}$. In its usual notation the bound has the shape

$$\sum_{\text{exc}} X^{2\sigma_j} \left| \sum_{n \asymp N_0} a_n \rho_j(n) \right|^2 \ll (1 + \sqrt{\mu N_0 X})(1 + \sqrt{\mu N_0^{1+\epsilon}}) \sum_n |a_n|^2,$$

with $\mu \asymp 1/M$, $M = 4dq^2$. The first factor is of size $1 + (F'/(q\sqrt{d}))^{1/2}$, which is not uniformly harmless near the boundary. The level average in [DI, Theorem 6] offers further partial control. The source calculation reports an extra factor $(F')^{3/4} D^{1/8}/A$ there; its detailed bookkeeping was not re-checked in the internal review, and we do not use it as a proved obstruction theorem. The same caveat applies to the source’s exploratory exceptional-density calculations. None enters the conditional proof.

A positive-width region bounded only by the original $1/j$ saving still contributes at the level of the available upper bounds $N\mathcal{L}^2 \sum_{j \ll \mathcal{L}} 1/j$. Unlike the linearly degenerating saving in Lemma 8.1, this does not remove $\log \log N$. With single-level estimates the strip has fixed width, so they give no improvement of [ET, Theorem 1.1]. Section 9 replaces them by the level-averaged bound [DI, Theorem 7] for $a_n = 1$ (the partial-control level average [DI, Theorem 6], for general a_n , does not suffice). The strip then shrinks to $\delta < w$ for every fixed $w > 0$, which gives Theorem 2, and to $w \asymp 1/\log \log N$ with the effective constants of Appendix A, which gives Theorem 1.

10.2. Lifts versus equidistribution of a finite packet. The lift-counting target (H^{**}) of [MN3, §3.5] is stronger than what the proof above uses. For the packets of natural class-number size $d^{1/2+o(1)}$ on a surface of volume $d^{1+o(1)}$, a box of hyperbolic area V has expected packet count $Vd^{-1/2+o(1)}$. Consequently a uniform relative equidistribution statement in *all* boxes cannot extend below area $d^{1/2-o(1)}$: integrality and boxes containing a packet point already prevent it. Here “scale” is area, not hyperbolic radius. This elementary granularity obstruction does *not* refute (H^{**}) as formulated for lifts, averaged in d , with congruence conditions. Nor does it obstruct the second-moment method here: the lift-counting Poincaré function has box area $\asymp A\sqrt{d}$, and Proposition 4.3 uses only its variance against a separated packet.

For comparison, Liu–Masri–Young [LMY, Theorem 1.4, and the more general Theorem 7.1] prove a joint level–discriminant equidistribution result in the range $q \leq |D|^{1/20-\epsilon}$. Their q is a prime level, $-D$ is an odd fundamental discriminant, and q splits in $\mathbb{Q}(\sqrt{-D})$. Their count is in translates of a level-one fundamental domain; it is not the present thin-box count with sieve twists. Here level and discriminant are tied, already d and $-4d$ before inserting q , and the coprimality/splitting hypotheses do not match. Thus that theorem does not supply (H^{**}) or the needed per- d estimate. We make no exhaustive literature claim that no other joint-aspect theorem exists.

11. OPEN PROBLEMS

Problem 1 (verification and numerical constants). Verify Appendix A independently (it has had internal review only), and make A_0 numerically explicit: re-derive

the fixed absolute majorants that were only shape-checked (listed at the end of Appendix A) and certify absolute values of C_W, B_W in the twisted Weil bound (B3). A cleaner route would be a published effective form of [DI, Theorem 7] and [Dr, Lemma 4.10], or any loss $\exp(o(\log(QN)/\log\log(QN)))$ in place of $(QN)^\varepsilon$, which suffices by Theorem 9.9(iii).

Problem 2 (asymptotic and other routes). Theorem 1 and [ET, Theorem 1.1] give $\sum_{p \leq N} f_I(p) \asymp N \log^2 N$. Decide whether $\sum_{p \leq N} f_I(p) \sim c_I N \log^2 N$ for some constant c_I ; the bad region of Section 2 is bounded here only up to constants. Independently, a proof that avoids the level average over d would be of interest, e.g. an exceptional large sieve for the box coefficients at a single level with losses $N^{o(1/\log\log N)}$.

Problem 3 (square-modulus Kloosterman sums). Obtain useful cancellation in $\sum_{a \asymp A} S(h, k; a^2)$, or its $4a^2$ variant, uniformly in the Fourier ranges produced by (7.3). For the application the averaging must handle the accompanying smooth weights, gcd factors, and the sieve congruence. Termwise square-root cancellation is exactly the estimate that stops at $D \asymp A$.

Problem 4 (joint level–discriminant counting). Establish a version of the lift-counting target (H**) at level comparable to the discriminant, with the parity restriction and congruence twists retained. A second-moment theorem for the appropriate Poincaré functions could suffice and would avoid a false assertion of uniform fine-scale equidistribution of a finite packet.

Problem 5 (general numerators). Extend the conditional estimate (see Remark 9.12 for the (SEL_m) version of [MN4]) and the unconditional estimate of Section 9 uniformly from $4/n$ to m/n . Primes dividing m must be omitted from the sieve, and the mass gain $\varphi(m)/m$ must be tracked against the sieve-product inflation $m/\varphi(m)$ and all remainder terms [MN3, remarks after Theorem 3.8]. The earlier internal transition gap $(m \log m/\varphi(m))^{1/3}$ is historical: the refined lower-side bound in [MN3, §2, Theorem L'] reduces it to $(\log\log N)^{1/3}$. An m -uniform removal of the Type I logarithmic loss would close that remaining gap at the scale $\log N \asymp m^{1/3}$; [MN4] does this under (SEL_m). An m -uniform version without (SEL_m) is not proved here: it would need the level average of Section 9 for the levels mdq^2 together with the effective constants of Appendix A.

Remark 11.1 (numerical status: EVIDENCE). The script `scripts/ttl_perd.py` supplies only a weak sanity check for per- d counts. Its weights are normalized to mass about 20, and its local-density comparison is not the orbital main term of Theorem 6.2. Small observed errors are therefore neither evidence for a uniform spectral error bound at the required scales nor a test of (SEL). No numerical observation is used in any proof.

APPENDIX A. EFFECTIVE VERSIONS OF DESHOULLERS–IWANIEC THEOREM 7 AND DRAPPEAU LEMMA 4.10

This appendix is a write-up of the internal note [TTL3]. Its long computations are in derivation files to which we point precisely; here we give the statement, the black boxes, the structure of the effective induction (with complete proofs of the inductive steps, including the small- C branch needed for nebensystem), the effective forms of the three inputs, and the corrections to [DI, Dr] found along the way.

Status: PROVED relative to the black boxes (B1)–(B5) below and to the derivations in [TTL3]; internally reviewed (three internal referee reports and two independent deep reviews, no fatal or major findings), not externally refereed. It needs expert scrutiny.

A.1. Statement. Fix $q_0 \geq 1$ and an even character χ modulo q_0 . For a level M with $q_0 \mid M$ the exceptional forms u_j of $(\Gamma_0(M), \chi)$ have $t_j = i\sigma_j$, $0 < \sigma_j \leq 1/4$, and DI-normalised coefficients $\rho_{j\infty}(n)$ (Drappeau’s ρ_f convert by $\rho^{\text{DI}}(n) = 2n^{1/2}\rho^{\text{Dr}}(n)$, Section 9). For $Q > 0$, $Y, N \geq 1$, $t \in \mathbb{R}$ and a closed interval $I = [N, N_1]$, $N \leq N_1 \leq 2N$, put

$$S_\chi(Q, Y, N, t; I) = \sum_{Q < q \leq 16Q} \sum_{j \text{ exc } (q_0 q, \chi)} Y^{2\sigma_j} \left| \sum_{n \in I} n^{it} \rho_{j\infty}(n) \right|^2,$$

and $S_\chi^*(Q, Y, N) = \sup_I S_\chi(Q, Y, N, 0; I)$. (The level is $M = q_0 q$, following [Dr, p. 17]; for $q_0 = 1$, $\chi = 1$ this is DI’s sum [DI, (8.4)] and we write S, S^* .)

Theorem A.1 ((EFF); PROVED relative to (B1)–(B5) and [TTL3]). *There is an absolute A_0 such that Corollary 9.1, for every $q_0 \geq 1$ and every even character χ modulo q_0 , holds with*

$$C_\varepsilon \leq 200 \varepsilon^{-1} K_7^\chi(\varepsilon/6) \leq \exp(\exp(A_0/\varepsilon)) \quad (0 < \varepsilon \leq 1/4),$$

where K_7^χ is the constant of Theorem A.4 below. One may take $A_0 = 24B_\chi + 30$, $B_\chi = 400 + 64b + 16 \log(2 + C_W)$, $b = \max(4, \lceil B_W + 1 \rceil)$, with C_W, B_W from (B3); e.g. $A_0 < 2 \cdot 10^4$ if $B_W \leq 3$, $C_W \leq 10$. This is Theorem 9.8 of Section 9.

Black boxes. The following are used as exact identities or as published theorems with absolute constants; no ε enters them.

- (B1) The Petersson and Kuznetsov trace formulae for $\Gamma_0(q)$ at the cusp ∞ [DI, §4, (1.19)], for $(\Gamma_0(M), \chi)$ at ∞ [Dr, Lemma 4.5, (4.13)–(4.17)] (from Blomer–Harcos–Michel), and for $\text{SL}_2(\mathbb{Z})$.
- (B2) Weil’s bound $|S(m, n; c)| \leq \tau(c)(m, n, c)^{1/2} c^{1/2}$.
- (B3) A twisted Weil bound $|S_\chi(m, n; c)| \leq C_W \tau(c)^{B_W} (m, n, c)^{1/2} (cq_0)^{1/2}$ for χ modulo $q_0 \mid c$, with absolute C_W, B_W [Dr, Lemma 4.2]. Their numerical values are not certified here; any absolute values suffice for the form $\exp(\exp(A_0/\varepsilon))$, and only the numerical A_0 depends on them. ($C_W = B_W = 1$ was checked for $c \leq 90$: EVIDENCE only.)
- (B4) Selberg’s $\lambda_1 \geq 3/16$ for $\Gamma_0(q)$ and $\Gamma_1(M)$, i.e. $\sigma_j \leq 1/4$.
- (B5) $\lambda_1(\text{SL}_2(\mathbb{Z})) > 1/4$ [DI, Theorem 3].

A.2. Effective toolkit. A constant $K(\delta)$, $0 < \delta \leq 1/4$, is of class $\mathcal{E}(B)$ if $K(\delta) \leq \exp(\exp(B/\delta))$. The ε -losses in [DI, Dr] come from divisor bounds, logarithms, integrations by parts of growing order, and the inductions on Q . The first three are handled by three elementary lemmas [TTL3, §1].

Lemma A.2 (PROVED). (a) *If f is multiplicative with $0 \leq f(p^a) \leq (a+1)^B$ ($B \geq 1$), then $f(n) \leq (2B/\delta)^{B \cdot 2^{B/\delta}} n^\delta$ for $0 < \delta \leq 1$ (class $\mathcal{E}(B \log 2 + o(1))$).* (b) *$(\log x)^r \leq (r/(e\delta))^r x^\delta$ for $x \geq 1$, $r, \delta > 0$.* (c) *There is $\eta \in C^\infty(\mathbb{R})$ with $0 \leq \eta \leq 1$, $\eta = 1$ on $[1, 2]$, $\text{supp } \eta \subset [3/4, 9/4]$ and $\|\eta^{(p)}\|_\infty \leq 8e^{16} 36^p (p!)^2$ for all $p \geq 0$ (Gevrey class 2).*

Proof. (a) $f(n)n^{-\delta} \leq \prod_{p^a \parallel n} ((a+1)p^{-a\delta/B})^B$; factors with $p \geq 2^{B/\delta}$ are ≤ 1 , and for the at most $2^{B/\delta}$ others $\max_a (a+1)2^{-a\delta/B} \leq 2B/\delta$. (b) Maximise $r \log u - \delta u$. (c) $h(x) = e^{-1/x}$ ($x > 0$) has $|h^{(p)}| \leq 9^p (p!)^2$ by Cauchy's formula on $|z-x| = x/2$; mollify $\mathbf{1}_{[7/8, 17/8]}$ with the rescaled normalised bump $h(1/4+x)h(1/4-x)$, whose mass is $\geq e^{-16}/4$, and use Leibniz. $\square$

With η from (c) in place of DI's unspecified smooth cutoffs, an integration by parts of order $p \asymp 1/\delta$ costs $\exp(O(\delta^{-1} \log(2/\delta)))$, well inside $\mathcal{E}(\cdot)$; the divisor bounds (a) are what produce the double exponential.

A.3. The effective induction. Two trivial facts hold for every q_0, χ [TTL3, §2]:

(M) for $Y \geq Y' \geq 1$, since $0 < 2\sigma_j \leq 1/2$ by (B4),

$$S_\chi(Q, Y', N, t; I) \leq S_\chi(Q, Y, N, t; I) \leq (Y/Y')^{1/2} S_\chi(Q, Y', N, t; I);$$

(PS) $S_\chi(Q, Y, N, t; I) \leq 2(1+t^2)S_\chi^*(Q, Y, N)$, by Abel summation over the admissible intervals $[N, \xi]$ and Cauchy–Schwarz for $d\xi/\xi$.

The non-inductive inputs are, for $\delta \in (0, 1/10]$, $Y, N \geq 1$ and all I , with constants $K_1, c, K_2 \geq 1$ independent of q_0, χ :

(P1) ([DI, (8.7) and Theorem 14]) for all $Q > 0$: $S_\chi^*(Q, Y, N) \leq K_1(\delta)(NY)^\delta(Q + N + Y)N$;

(P2) ([DI, Lemma 8.1], [Dr, pp. 17–18]) for $Q \geq 1$, with $C := \pi NY/(q_0 Q)$: $S_\chi(Q, Y, N, 0; I) \leq c(\delta) \int_{\mathbb{R}} S_\chi(C, Y, N, t; I) \frac{dt}{1+t^4} + c(\delta)(QYN)^\delta(Q + N + NY/Q)N$;

(P3) ([DI, Theorem 2, (1.29)], summed over $Q < q \leq 16Q$) for $Q \geq 1$: $S_\chi(Q, 1, N, 0; I) \leq K_2(\delta)(Q + N^{1+\delta})N$.

In (P2) the sum $S_\chi(C, \dots)$ is defined for every $C > 0$ (it is empty if $16C < 1$). For $q_0 = 1$ the switched parameter is $C = \pi NY/Q$, and the induction below only uses (P2) when $C \geq N$. For $q_0 > 1$ the factor $1/q_0$ can make $C < N$, even $C < 1$, where no induction hypothesis is available; this is the extra branch (C2) below. (Drappeau's remark that q_0 “appears only with negative powers” needs this branch.)

Proposition A.3 (effective [DI, (8.19)], with nebensystem; PROVED relative to (P1)–(P3)). *Let $\delta \in (0, 1/10]$, $Q_0 := \max((90c)^{1/(10\delta^2)}, (2\pi)^{1/(2\delta)})$ and $H := \max(2K_2Q_0, 10K_1, 200cK_1)$. Then for all $Q \geq 1$ and $1 \leq N \leq Q$,*

$$S_\chi^*(Q, Q^{2-2\delta}/N, N) \leq HQ^{1+4\delta}N.$$

Proof. Put $Y = Q^{2-2\delta}/N \geq 1$; induct on k with the claim for $Q \leq 2^k Q_0$. (A) $Q \leq Q_0$: by (M) with $Y' = 1$ and (P3), $S_\chi^* \leq Y^{1/2} K_2(Q + N^{1+\delta})N \leq Q^{1-\delta} N^{-1/2} \cdot 2K_2 Q^{1+\delta} N \leq 2K_2 Q_0 Q^{1+4\delta} N$. (B) $Q^{1-2\delta} < N \leq Q$: by (M) with $Y_1 = Q + N \geq Y$ and (P1), $S_\chi^* \leq K_1(2Q^2)^\delta \cdot 4QN \leq 10K_1 Q^{1+4\delta} N$. (C) $Q > Q_0$, $N \leq Q^{1-2\delta}$: then $Q^{2\delta} \geq 2\pi$ and $C = \pi Q^{1-2\delta}/q_0 \leq Q/2$; the error term of (P2) is $\leq 3cQ^{1+3\delta}N$. (C1) $C \geq N$. With $Y_C = C^{2-2\delta}/N \leq Y$, the induction hypothesis at (C, N) , (M) and (PS) bound the main term of (P2) by

$$c \int \left(\frac{Y}{Y_C}\right)^{1/2} 2(1+t^2) H C^{1+4\delta} N \frac{dt}{1+t^4} = 2\sqrt{2}\pi c H Q^{1-\delta} C^{5\delta} N,$$

which is $\leq 45c H Q^{1+4\delta-10\delta^2} N \leq \frac{H}{2} Q^{1+4\delta} N$, using $\int (1+t^2)dt/(1+t^4) = \sqrt{2}\pi$, $C^{5\delta} \leq \pi^{5\delta} Q^{5\delta-10\delta^2}$ and $Q^{10\delta^2} \geq 90c$. (C2) $C < N$. No induction: (M) with $Y_1 = C+N \in [1, 2N]$ and (P1) at C give $S_\chi^*(C, Y, N) \leq (1+\sqrt{Y/N})K_1(2N^2)^\delta 4N^2 \leq$

$10K_1Q^{1+2\delta}N$ (as $\sqrt{YN} = Q^{1-\delta}$), so the main term is $\leq 90cK_1Q^{1+2\delta}N$. In both cases $S_\chi(Q, Y, N, 0; I) \leq (H/2 + 90cK_1 + 3c)Q^{1+4\delta}N \leq HQ^{1+4\delta}N$; take the supremum over I . $\square$

For $q_0 = 1$ branch (C2) never occurs and one may take $H = \max(2K_2Q_0, 10K_1, 6c)$ [TTL3, Proposition 2.1]. Removing the restrictions $N \leq Q$, $Y = Q^{2-2\delta}/N$ by (M) and (P1) gives:

Theorem A.4 (effective [DI, Theorem 7], [Dr, Lemma 4.10]; PROVED relative to (P1)–(P3)). *For $0 < \delta \leq 1/10$, $Q \geq 1/16$ and $Y, N \geq 1$, uniformly in q_0 and even χ ,*

$$S_\chi^*(Q, Y, N) \leq K_7^\chi(\delta)(QN + 1)^{5\delta}(Q + N + \sqrt{NY})N, \quad K_7^\chi := \max(H, 10K_1).$$

If $K_1, c, K_2 \leq \exp(\exp(B/\delta))$ with $B \geq 1$, then $K_7^\chi(\delta) \leq \exp(\exp((B+3)/\delta))$.

Proof. $Q \geq 1$, $N \leq Q$: if $Y \leq Y_2 := Q^{2-2\delta}/N$ use (M) and Proposition A.3; otherwise $S_\chi^* \leq (Y/Y_2)^{1/2}HQ^{1+4\delta}N = HQ^{5\delta}\sqrt{NY}N$. $N > Q$ or $Q < 1$: (M) with $Y_1 = Q + N \leq 2N$ (resp. $1 + N$) and (P1) give $\leq (1 + \sqrt{Y/N})K_1(2N^2)^\delta 4N^2 \leq 10K_1N^{2\delta}(N + \sqrt{NY})N$. Growth: $\log Q_0 \leq (\log 90 + e^{B/\delta})/(10\delta^2) + \delta^{-1}$, and $\delta^{-2} \leq e^{2/\delta}/4$, $e^{B/\delta} \geq e^{10}$ [TTL3, Corollary 2.3]. $\square$

A.4. The three inputs. Each input is a re-derivation of a printed proof in which every ε -loss is replaced by an explicit constant; the full computations are in the derivation files named below.

Proposition A.5 (effective [DI, Theorem 2] at ∞ ; PROVED relative to (B1), (B2), (B4)). *For $q, K \geq 1$, $N \geq 1/2$, $0 < \delta \leq 1/10$, each of [DI, (1.28)–(1.30)] (exceptional spectrum included) at the cusp ∞ of $\Gamma_0(q)$ is $\leq K_{T_2}(\delta)(K^2 + q^{-1}N^{1+\delta})\|a\|^2$, with $K_{T_2}(\delta) \leq \exp(\exp(100/\delta))$. With nebentypus χ modulo q_0 at level $M = q_0q$, using (B3) instead of (B2), the same holds with a constant $K_{LS}^\chi(\delta) \leq \exp(\exp(B_\chi/\delta))$ independent of q_0, χ .*

Source. `scripts/ttl3_thm2_effective.md` (Steps 1–9) and `scripts/ttl3_thm2_twisted.md`. The only growing-order step is [DI, Proposition 3, (1.27), p. 257]: with η of Lemma A.2(c) and $p = \lfloor 32/\delta \rfloor$ integrations by parts the non-resonant Poisson terms cost $2^{100(p+1)}(p!)^4$; the divisor factors $\tau(c)^{\leq 4}$ cost $(8/s)^{4 \cdot 2^{4/s}}$, $s = \delta/16$, which dominates. The diagonal is kept as $D_K \leq 2K^2$. In the twisted case only the Weil range $c > N^2$ acquires a factor $q_0^{1/2}$, which $M^{-1} = (q_0q)^{-1}$ absorbs; the diagonal is not multiplied by $q_0^{1/2}$. Then (P3) holds with $K_2 = 32K_{T_2}$ (resp. $32K_{LS}^\chi$): for exceptional κ_j , $\cosh(\pi\kappa_j) = \cos(\pi\sigma_j) \geq 2^{-1/2}$, $\|\mathbf{1}_I\|^2 \leq 2N$, and $\sum_{Q < q \leq 16Q} (1 + N^{1+\delta}/q) \leq 16(Q + N^{1+\delta})$.

Proposition A.6 (effective [DI, Theorem 14]; PROVED relative to (B1), (B5) and Proposition A.5 at level 1). *For $C, M, N \geq 1$, $0 < \delta \leq 1/10$,*

$$\sum_{c \leq C} U(c) \leq K_{14}(\delta)(CMN)^\delta C(C + MN),$$

where

$$U(c) := \sum_{d \bmod c}^* \left| \sum_{m \leq M} e\left(\frac{md}{c}\right) \right| \left| \sum_{n \leq N} e\left(\frac{nd}{c}\right) \right|,$$

with $K_{14}(\delta) = 2^{1200}(1 + K_{T_2}(\delta))(1 + 6/\delta)^3$.

Source. `scripts/ttl3_thm14_effective.md`. The quantity $U(c)$ dominates $|\sum_{m,n} S(m,n;c)|$ and the twisted sums $|\sum_{m,n} S_\chi(m,n;c)|$; it is what DI's proof via [DI, Lemma 8.2, (8.14)] actually bounds. Only fixed-order derivatives and fixed contour shifts occur; since $\mathrm{SL}_2(\mathbb{Z})$ has no exceptional spectrum (B5), no inductive or ineffective step enters.

Proposition A.7 (effective [DI, Lemma 8.1] and (P1); PROVED relative to (B1)–(B4), Propositions A.5, A.6). *(P1) and (P2) hold with $K_1(\delta) = A\delta^{-2}[1 + K_{T_2}(e) + D(e)K_{14}(e)]$ and $c(\delta) = A\delta^{-2}[1 + K_{T_2}(e)]$, where $e = \delta/4$, $D(e) = (2/e)^{2^{1/e}}$, $A = 2^{16384}$; in the twisted case K_{T_2} is replaced by K_{LS}^χ , giving $K_1^\chi, c^\chi \leq \exp(\exp((4B_\chi + 1)/\delta))$.*

Source. `scripts/ttl3_lemma81_effective.md` and `scripts/ttl3_lemma81_twisted.md`. Explicit cutoffs $\Psi(u) = \eta(3u - 2)$, $\varphi(x) = \Psi(Yx)$, $f(q) = \eta(q/Q)$ make the switched moduli satisfy $64C/51 \leq c \leq 128C/11$, inside $(C, 16C]$. The exceptional lower bound $\hat{\varphi}(-i\sigma)/\cos(\pi\sigma) \geq Y^{2\sigma}/64$ holds uniformly in $\sigma \in (0, 1/4]$ for $Y \geq 2^{32}$; smaller Y is handled by (M) and Proposition A.5. The Mellin pair for $f = \eta(\cdot/Q)$ satisfies $|\chi_f(it)| \leq 2^{512}/(1+t^4)$ uniformly. The exceptional spectrum is split at $\sigma = e$ (cost e^{-1} from $1/\sin \pi\sigma$, and $Y^{2e}(1 + \log Y)$). In (P1) the weight $\Psi(4\pi Y \sqrt{mn}/k)$ is removed by two-variable partial summation against a k -independent majorant, then $\tau(k) \leq D(e)k^e$ (Lemma A.2(a)) and Proposition A.6 with $K = 32NY$. If $\pi NY/(q_0 Q) < 1$ the error term alone suffices.

A.5. Proof of Theorem A.1. For $q_0 = 1$: $K_{T_2}(\delta/4) \leq \exp(\exp(400/\delta))$, $K_{14}(\delta/4) \leq \exp(\exp(404/\delta))$, $D(\delta/4) \leq \exp(\exp(3/\delta))$, so $K_1, c \leq \exp(\exp(405/\delta))$ and $K_7(\delta) \leq \exp(\exp(408/\delta))$ by Theorem A.4 [TTL3, Theorem 7.1]. In general $K_1^\chi, c^\chi \leq \exp(\exp((4B_\chi + 1)/\delta))$, $K_2^\chi = 32K_{LS}^\chi(\delta)$, hence $K_7^\chi(\delta) \leq \exp(\exp((4B_\chi + 4)/\delta))$. Conversion to Corollary 9.1: the levels $M = q_0 q \leq M_0$ are covered by $\leq 2 + \log M_0$ blocks $Q_i < q \leq 16Q_i$ with $Q_i \geq 1/16$, and $[1, t]$ by the $\lfloor \log_2 t \rfloor + 1 \leq 3 \log^2(2t)$ closed intervals $[2^k, \min(2^{k+1} - 1, t)]$; Cauchy–Schwarz over the latter, positivity over the former, and Theorem A.4 with $\delta = \varepsilon/6$ give, using Lemma A.2(b) for $(2M_0 t)^{5\delta}(2 + \log M_0) \leq (12/\varepsilon)(M_0 t)^\varepsilon$, the bound of Corollary 9.1 with $C_\varepsilon \leq 200\varepsilon^{-1}K_7^\chi(\varepsilon/6)$. Finally $\log \log(200\varepsilon^{-1}K_7^\chi(\varepsilon/6)) \leq (24B_\chi + 30)/\varepsilon$. $\square$

A.6. Corrections to the printed proofs. None of the following affects the theorems of [DI, Dr]; they are recorded because an effective version must use the repaired forms [TTL3, §§4–6, 8].

- (1) [DI, Lemma 8.1], [Dr, p. 17]: the pictured cutoff supports $[1/2, 5/2]$ give only $16C/25 \leq c \leq 32C$, not $C < c \leq 16C$; repaired by narrower cutoffs (above). In [Dr, p. 17] a single $g(q) = \Phi(q/Q)$ cannot majorise $Q < q \leq 16Q$; DI's four applications at $(2^j Q, 2^j Y)$, $j = 0, \dots, 3$, are needed.
- (2) [DI, (8.3)]: the exceptional lower bound is not uniform near $Y = 1$; [DI, (8.2)] needs an extra $\log Y$ at $\kappa = 0$ (both repaired above). The sign convention of [DI, (1.22)] is opposite to that of (8.1); both applications carry the same sign, which cancels.
- (3) [DI, p. 257]: the cutoff support must be checked for the derivative separation; [DI, p. 259]: the exponent $2s$ should be $3s$. The asymptotic diagonal [DI, (5.2)] is replaced by $D_K \leq 2K^2$; the Gaussian lower bound [DI, pp. 260–261] is restricted to $[|r|, |r| + 1]$; the kernel in [DI, (5.6)] has $\tanh \xi$; the

holomorphic kernel on p. 258 has the wrong overall sign (only its modulus is used).

- (4) [DI, Theorem 13]: the Fourier separation differentiates up to six times, beyond what [DI, (7.7)] controls (repaired by product cutoffs); a factor $\cosh$ is missing on p. 264.
- (5) [DI, (8.18)]: $Y_1 = \sqrt{Q+N}$ should read $Y_1 = Q + N$ (Section 9).
- (6) [Dr, pp. 17–18]: $S_{\infty\infty}(m, n; qc)$ should be $S_{\chi}(m, n; q_0 qc)$; the Mellin kernel needs dx/x , as in [Dr, (4.10)], not dx ; the second coefficient has index n , not m ; $S(Q, N, Y, 0)$ should be $S(Q, Y, N, 0)$. Drappeau’s pre-Kuznetsov kernel [Dr, p. 14] is $\cosh(\pi t)$ times DI’s (an absolute normalisation factor). The induction needs the small- C branch (C2) of Proposition A.3.

A.7. Status of the derivations. After the internal referee round [TTL3, referee reports] and two independent deep reviews, the following were independently re-derived: the toolkit, the induction and its nebentypus branch, the assembly and the conversion (all written out above); the ε -structure and the key inequalities of Propositions A.5–A.7 and of their twisted forms; and the corrections above. Some fixed absolute majorants were only shape-checked (asserted parameter-independent, not re-derived in detail): `ttl3_thm2_effective.md` Steps 6–8 (angular split of E_K , Gaussian lower bound 2^{-20}), `ttl3_thm14_effective.md` §§2–4 (Mellin–Barnes residues, the constants 2^{50} , 2^{1000}), and the numerals 2^{512} , 2^{2048} of `ttl3_lemma81_effective.md`. Hence the numerical value of A_0 is explicit only modulo these allowances and certified values of C_W, B_W ; the form $\exp(\exp(A_0/\varepsilon))$, and therefore Theorem 9.9(ii), need only their finiteness and independence of the parameters.

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